

Chapter 1

INTRODUCTION

1.1 Overview

The project on Forensic Face Sketch Construction and Recognition revolves around developing and applying technology for creating, analyzing, and recognizing forensic facial sketches, which are typically used in criminal investigations. These sketches are created by forensic artists based on witness descriptions of suspects and can be crucial for identifying and apprehending criminals. The core of this project involves two main aspects: Face Sketch Construction and Face Recognition. This part of the project focuses on automatically generating face sketches from textual or verbal descriptions provided by eyewitnesses. The process is often time-consuming and subjective, involving skilled forensic artists. However, using advanced computer vision techniques, machine learning, and deep learning. The second major component of the project focuses on the recognition of faces in photos, videos, or live surveillance footage using the generated forensic sketches. This helps law enforcement agencies match the sketches with real-world individuals or suspects

In criminal investigations, the identification of suspects often relies on eyewitness accounts. Traditionally, forensic artists are employed to create facial sketches based on witness descriptions, but this process is time-consuming, subjective, and highly dependent on the artist's skill. To address these limitations, this project “Third Eye: Forensic Face Sketch Construction and Recognition” presents a digital solution that enables law enforcement agencies to construct composite facial sketches and identify suspects using automated facial recognition technology.



The system provides an intuitive graphical user interface (GUI) where users can drag and drop predefined facial components such as eyes, noses, lips, and hairstyles to assemble a suspect's face based on verbal descriptions. Once the sketch is completed, it is processed and matched against a database of real facial images using cloud-based recognition services. The project integrates JavaFX for the user interface, AWS S3 for secure image storage, and AWS Recognition for facial similarity detection. This approach not only reduces reliance on forensic artists but also accelerates the suspect identification process and enhances accuracy by leveraging cloud computing and artificial intelligence. The system is especially useful in the early stages of criminal investigations, where rapid suspect identification can be critical. In the field of criminal justice, the ability to quickly and accurately identify suspects is critical. Traditionally, suspect identification based on eyewitness descriptions requires the assistance of skilled forensic artists who manually draw sketches. However, this process can be slow, subjective, and costly — with varying degrees of accuracy depending on the artist's interpretation and the witness's memory. This project, titled “Third Eye: Forensic Face Sketch Construction and Recognition,” presents a modern, technology-driven alternative. It enables users such as police officers or forensic personnel to construct composite facial sketches digitally using a library of predefined facial features. These sketches are created through an interactive and user-friendly drag-and-drop interface built using JavaFX.



Once the sketch is created, it is uploaded to the cloud (AWS S3) and compared against a database of real face images using AWS Recognition, a powerful facial recognition service. The system can return potential matches, helping investigators narrow down suspect lists more efficiently and without requiring artistic skills.

By integrating cloud computing, artificial intelligence, and a customizable graphical interface, this system provides a scalable and accessible solution for enhancing traditional sketch-based suspect identification. It not only speeds up the investigative process but also democratizes forensic sketch creation for use in a wide range of law enforcement scenarios. “Third Eye” is a forensic face sketch and recognition system designed to assist law enforcement in identifying suspects based on eyewitness descriptions. Instead of relying on hand-drawn sketches by forensic artists, this system offers a digital interface where users can assemble composite sketches using predefined facial features.



Once a composite sketch is built, the system uses AWS Recognition to compare it against a database of real faces and returns potential matches. The project leverages JavaFX for the interface and AWS for cloud-based storage and facial recognition, offering a fast, accessible, and cost-effective tool for modern policing. These sketches are then uploaded to the cloud and matched against a database of real facial images using AWS Recognition. By combining Java-based front-end development with cloud-based AI services, the system enhances the speed and accuracy of suspect identification, reducing reliance on traditional forensic sketch artists. In many real-world scenarios, especially in the early stages of an investigation, law enforcement agencies must rely on vague or incomplete eyewitness accounts to identify suspects. Often, these descriptions are not sufficient to generate accurate leads without professional sketch artists, who may not always be available. The “Third Eye” system bridges this gap by digitizing the facial sketching process and integrating it with cloud-based recognition technology. By combining digital sketch creation tools with AI-based facial recognition, the Third Eye system provides a faster, more consistent, and scalable solution for suspect identification. While effective, this process is time-consuming, subjective, and depends heavily on the skill of the artist. It reduces reliance on manual artistry, enhances recognition accuracy, and brings greater efficiency to forensic investigations.

1.2 Problem Statement

In many criminal investigations, creating an accurate composite sketch of a suspect is a time-consuming and skill-dependent task. Traditional methods rely heavily on forensic artists, who may not always be available, and the process can be subjective and inefficient. Additionally, matching these sketches with existing police databases is often manual and slow, delaying critical investigations.

We present a standalone application that allows users without requiring forensic expertise to create composite facial sketches of suspects using a simple drag-and-drop interface. The application leverages deep learning algorithms to automatically and efficiently match the created sketches with images in police databases.

By utilizing cloud infrastructure, the system ensures scalability, faster processing, and remote accessibility, significantly enhancing the speed and accuracy of criminal identification. In many criminal investigations, creating an accurate composite sketch of a suspect is a time-consuming and skill-dependent task. Traditional methods rely heavily on forensic artists, who may not always be available, and the process can be subjective and inefficient.

Additionally, matching these sketches with existing police databases is often manual and slow, delaying critical investigations. We present a standalone application that allows users without requiring forensic expertise to create composite facial sketches of suspects using a simple drag-and-drop interface. The application leverages deep learning algorithms to automatically and efficiently match the created sketches with images in police databases.

By utilizing cloud infrastructure, the system ensures scalability, faster processing, and remote accessibility, significantly enhancing the speed and accuracy of criminal identification. In traditional criminal investigations, suspect identification often relies heavily on eyewitness accounts and the manual sketching of faces by trained forensic artists.

However, this process is inherently slow, subjective, and dependent on the availability and skill of the artist. In many cases, the resulting sketches may lack accuracy due to limitations in memory recall or artistic interpretation, leading to delays or errors in identifying suspects.

Moreover, matching hand-drawn sketches to real photographic records in criminal databases is a non-trivial task that lacks automation in most law enforcement systems. There is a clear need for a system that can digitally construct accurate facial composites based on verbal descriptions and automatically match these composites against a database of real faces.

1.3 Solution

To address the limitations of traditional forensic sketching and manual suspect identification, the Third Eye project offers a comprehensive digital solution that integrates facial sketch construction with automated face recognition. The system features an intuitive graphical user interface (GUI) where users such as police officers or forensic investigators can build composite sketches by selecting and assembling predefined facial components like eyes, noses, lips, and hairstyles.

This eliminates the need for a professional sketch artist and ensures faster, more consistent sketch creation. Once a composite face is constructed, the system leverages cloud-based artificial intelligence, specifically Amazon Web Services (AWS) Recognition, to compare the sketch against a database of real facial images stored in AWS S3. The recognition system then returns potential matches ranked by similarity, significantly improving the speed and accuracy of suspect identification. This end-to-end solution streamlines the forensic process, reduces human error, and enhances the effectiveness of criminal investigations.

1.4 Existing System

In the current forensic investigation process, suspect identification often relies on hand-drawn sketches created by trained forensic artists based on eyewitness descriptions. This method is highly subjective, as it depends on the memory and verbal communication of the witness, as well as the skill of the artist. Additionally, the process is time-consuming and cannot be easily scaled, especially in cases involving multiple suspects or large regions. Once a sketch is drawn, matching it with real faces from criminal databases is typically done manually by investigators, which is both inefficient and prone to human error.

1.5 Proposed Systems

The Third Eye system introduces a digitized, automated approach to forensic face sketching and recognition. Instead of relying on artists, users can build a suspect's facial sketch using a drag-and-drop interface that provides a library of predefined facial features (eyes, noses, mouths, hairstyles, etc.). This enables consistent and rapid sketch creation by non-experts. The system then utilizes AWS Recognition, a cloud-based facial recognition service, to automatically match the constructed sketch with photographs stored in an AWS S3 database. Potential matches are returned based on facial similarity, significantly improving identification speed and accuracy. This system minimizes human bias, enhances efficiency, and provides a scalable, intelligent platform for law enforcement and forensic applications.

1.6 Objectives and scope

- **To create an easy-to-use application:** Develop an application where users can make a suspect's face by dragging and dropping different facial features (like eyes, nose, and mouth), without needing help from a professional forensic artist.
- **To match the composite sketch with real photos using deep learning:** Use deep learning (AI) techniques to compare the drawn face sketch with photos stored in the police database and find possible matches.
- **To make the system fast and scalable by using cloud infrastructure:** Use cloud services to store data and run the matching system, so the application works quickly and can handle a large number of cases at the same time.
- **To reduce time and dependency on forensic artists:** Help police and investigators create suspect face sketches faster and without waiting for a forensic artist.
- **To keep all data safe and secure:** Protect sensitive information, like suspect sketches and police records, by using secure cloud technologies

The scope of this survey is to explore and evaluate existing techniques, methodologies, and tools used for face sketch construction and sketch-based face recognition. The focus is primarily on understanding how current systems handle the process of creating, processing, and matching facial sketches with photographic databases to identify suspects.

This survey includes a review of:

- Traditional face sketch construction techniques, including manual forensic sketching and operator-assisted composite sketch tools.
- Modern approaches using machine learning, computer vision, and image synthesis techniques for converting hand-drawn or composite sketches into digital formats suitable for matching.
- Various recognition models such as those based on Markov Random Fields, SIFT Descriptors, and feature transformation algorithms.
- Systems capable of sketch-to-photo synthesis and matching, especially in cases where sketch/photo orientations differ or sketches have varying artistic styles.
- Existing limitations in terms of accuracy, usability, feature realism, and time-efficiency in earlier models.

The survey aims to identify the gaps in current research and systems, which include:

- Inability to effectively process and utilize hand-drawn individual facial features.
- Low recognition accuracy when sketch and photo orientations do not match.
- Inefficiencies in user experience and composite creation time in earlier systems.

By highlighting these gaps, the survey establishes a foundation for the proposed application, which introduces a more adaptive and intelligent platform for both sketch creation and suspect recognition. The new system aims to support law enforcement agencies with better tools for criminal identification by incorporating deep learning, user-uploaded hand-drawn features, and a scalable cloud-based infrastructure. The goal of the survey is to identify gaps in current systems and establish the need for an integrated, automated solution that improves both the speed and accuracy of suspect identification.

Chapter 2

LITERATURE SURVEY

2.1 Survey Papers

[1] K. K. Jain, S. Grosz, A. M. Namboodiri, and A. K. Jain, “CLIP4Sketch: Enhancing Sketch to Mugshot Matching through Dataset Augmentation using Diffusion Models,” 2024.

The paper titled “CLIP4Sketch: Enhancing Sketch to Mugshot Matching through Dataset Augmentation using Diffusion Models” by Kushal Kumar Jain, Steven Grosz, Anoop M. Namboodiri, and Anil K. Jain, presented in arXiv (2024), proposes a novel architecture for forensic sketch-to-mugshot face recognition. The core architecture, CLIP4Sketch, integrates latent diffusion models with CLIP and Ada Face embeddings to control both identity and sketch style during image generation. Techniques such as ControlNet for spatial guidance and textual prompts for style variation are used to generate diverse, realistic sketch datasets. The methodology involves creating synthetic sketches from mugshots using a diffusion-based pipeline conditioned on combined embeddings, enabling training of face recognition models in data-scarce sketch domains. Despite substantial improvements in sketch-to-photo matching, the approach faces limitations in handling highly distorted or subjective forensic sketches, particularly from datasets like IIIT-D and TUFTS.

Using this pipeline, the authors create a large-scale synthetic dataset comprising over 245,000 sketches across 27,000 identities in various styles. This synthetic data is then used to fine-tune a face recognition model for sketch-to-mugshot matching. Experimental evaluations demonstrate that the generated sketches not only resemble real forensic sketches in appearance and latent embedding space but also significantly enhance recognition accuracy across benchmark datasets such as Wild Sketch, IIIT-D, and TUFTS. Comparisons with previous GAN-based augmentation methods show that CLIP4Sketch outperforms them in both closed-set and open-set scenarios, achieving better Rank-1 identification rates and lower false negative rates. However, the authors also note that excessive reliance on synthetic data can slightly degrade performance on traditional face-to-face matching tasks, indicating a trade-off that must be carefully managed. In conclusion, CLIP4Sketch provides a powerful tool for augmenting training datasets in forensic face recognition tasks by generating high-quality, identity-aware sketches. It helps reduce the dependency on scarce real-world forensic data while also improving model generalization across sketch styles. The paper suggests future work could focus on better handling extreme sketch distortions and exploring similar augmentation strategies in other cross-modal recognition domains.

[2] N. M. Farid, M. S. Fard, and A. Nick Abadi, “Face Sketch to Photo Translation Using Generative Adversarial Networks,”

The paper titled “Face Sketch to Photo Translation Using Generative Adversarial Networks” by Nastaran Moradzadeh Farid, Maryam Saeedi Fard, and Ahmad Nickabadi, published as a preprint on arXiv in 2021, presents a GAN-based framework for converting facial sketches into realistic colored photos without requiring paired training data. The architecture integrates a pre-trained StyleGAN as a face generator, a neural network (F2W) for mapping sketch features to latent vectors, and an optimization module to refine image realism and identity fidelity. Key techniques include high-level feature extraction via VGGFace and manifold preservation using a face neness-scoring CNN. The methodology combines initialization and iterative refinement to generate high-quality outputs, demonstrating superior SSIM and rank-1 recognition scores compared to existing GAN-based models. A primary limitation is the higher inference time per image (3–5 minutes) due to iterative optimization, contrasting with the faster inference of encoder-decoder models.

Converting a black and white facial sketch into a photorealistic colored image is inherently challenging because sketches lack color, texture, and many fine details present in photographs. The authors propose a novel GAN-based framework that does not require paired sketch–photo datasets for training, thereby overcoming the scarcity of aligned data.

Their method consists of two major steps. In the initialization phase, a sketch is mapped into the latent space of a pre-trained face image generator (e.g. StyleGAN) via a learned mapping network (called F2W). This gives a coarse face photo prediction from the sketch. In the optimization phase, they iteratively adjust the latent vector to improve fidelity, guided by a composite loss that enforces similarity of high-level facial features between the sketch and generated image (using multiple feature extractors) and constrains the result to lie on a “manifold” of artifact-free face images using a learned face neness score network (HOGFD). The optimization refines texture, color, and finer structural consistency. Experiments on datasets like CUFS and AR show that the proposed method yields higher SSIM, FSIM, and VIF scores compared to Cycle GAN, DualGAN, and other baselines, and achieves a rank-1 recognition rate of 97.59 % in face matching tasks. Qualitatively, user studies also favor the generated photos for realism and identity consistency. The authors highlight that their method generalizes to unseen sketches, does not require retraining per dataset, but the per-image optimization is computationally more intensive. They suggest future directions like incorporating attribute input (e.g. hair/skin color) to further guide synthesis. In addition to addressing the lack of paired training data, the authors emphasize the flexibility and robustness of their approach in handling diverse sketch styles and levels of abstraction.

[3] W. Wan, Y. Yang, and H. J. Lee, “Generative adversarial learning for detail-preserving face sketch synthesis,” *Neurocomputing*, vol. 438, pp. 107–121

The paper titled “Generative Adversarial Learning for Detail-Preserving Face Sketch Synthesis” by Weiguo Wan, Yong Yang, and Hyo Jong Lee, published in *Neurocomputing*, Volume 438, Pages 107–121, in 2021, introduces a GAN-based framework tailored for preserving fine facial details in face sketch synthesis. The architecture is based on a modified high-resolution network acting as the generator, which gradually merges multi-resolution features to retain detailed information, coupled with a CNN discriminator.

The key techniques include a Laplacian of Gaussian (LoG)-based detail loss to enforce detail preservation, a style loss derived from pseudo sketch features using VGG19, and a total variation loss to reduce noise. The methodology encompasses training the network with adversarial, detail, style, and smoothness constraints on paired datasets (CUFS and CUFSF), achieving superior FSIM and Scoot scores. Limitations include challenges in synthesizing sketches for non-frontal faces and reliance on paired data, suggesting the need for more robust unsupervised or semi-supervised learning strategies.

This paper addresses limitations in existing face sketch synthesis methods, which often capture the overall facial content but fail to preserve fine details such as texture, small contours, and features (e.g. teeth, skin detail) when converting from photo to sketch. To solve this, the authors propose a generative adversarial network (GAN)-based framework that emphasizes detail preservation and style realism. They adapt a High-Resolution Network (HRNet) as the generator so that high-resolution representations are maintained at multiple scales, enabling finer features in the generated sketches. Beyond the usual adversarial loss (ensuring the outputs look sketch-like), they introduce a detail loss computed via the Laplacian of Gaussian (LoG) filter between the input photo and the generated sketch; this forces the network to produce edges and textures commensurate with the photo’s details. They also add a style loss to ensure the synthesized sketches respect the vivid, hand-drawn sketch style (e.g. in strokes, shading) seen in real sketch datasets. Experimental results on benchmark datasets (CUFS, CUFSF) demonstrate that their method yields higher FSIM and Scoot scores (indicating both perceptual similarity and style consistency) compared to prior methods; visually, their results show sharper textures and fewer blur or artifact issues. The work thus makes a strong contribution to sketch synthesis by balancing the trade-off between content fidelity and detail/style realism. In addition to the architectural innovations, the authors highlight the importance of balancing multiple loss functions to achieve both structural accuracy and stylistic realism in the generated sketches.

[4] L. Cao, J. Yin, Y. Guo, K. Du, and F. Zhang, “Sketch face recognition based on light semantic Transformer network,” *IET Comput. Vis.*, vol. 17, pp. 962–976, 2023

The paper titled “Sketch Face Recognition Based on Light Semantic Transformer Network” by Lin Cao, Jianqiang Yin, Yanan Guo, Kangning Du, and Fan Zhang, published in *IET Computer Vision*, Volume 17, pp. 962–976, in 2023, proposes a lightweight semantic Transformer-based architecture (LWVT-ResNet18) to enhance sketch face recognition (SFR).

The architecture integrates convolutional layers from ResNet18 with lightweight visual Transformer (LWVT) modules incorporating hierarchical group linear transformation (HGLT), tokenisers, and parameter sharing to reduce model complexity. Techniques include meta-learning for small sample training and a domain-adaptive focal loss to bridge modality gaps between sketches and photos. The methodology combines CNN-based feature extraction with Transformer modules to model consistent semantic features across modalities. The system shows notable performance improvements on CUFSF and UoM-SGFSv2 datasets. A key limitation lies in its complexity, requiring careful tuning and computational resources to maintain performance across datasets of varying styles and sample sizes.

The paper targets the challenging problem of sketch-to-photo face recognition, where cross-modality differences (sketch vs photograph) and limited training samples make discriminative feature learning difficult. To address this, the authors propose a light semantic Transformer architecture—called LWVT-ResNet18—that combines convolutional feature extraction with Transformer-style semantic modeling, but is specially designed to work on small datasets.

They integrate a meta-learning training strategy to form task-specific training batches (i.e. few-shot like) to mitigate overfitting in small sample regimes. The Transformer modules are “lightened” by introducing a hierarchical group linear transformation (HGLT) that expands and compresses token dimensions in stages, and by parameter sharing across Transformer layers, thereby reducing model complexity while retaining expressive power.

To further bridge modality gaps, they propose a domain-adaptive focal loss that emphasizes hard sample pairs and aligns sketch and photo embeddings more effectively. Experimentally, on benchmark sketch-photo datasets such as UoM-SGFSv2 and CUFSF, their method shows strong gains in recognition accuracy (e.g. achieving ~92.59 % on CUFSF) and clearly outperforms prior CNN- or domain-adaptation based methods. In addition to its strong performance, the proposed LWVT-ResNet18 architecture offers practical advantages in terms of computational efficiency and training scalability, making it suitable for real-world sketch face recognition applications where resources and data are limited.

[5] S. Bae, N. U. Din, H. Park, and J. Yi, “Face Photo-Sketch Recognition Using Bidirectional Collaborative Synthesis Network,”

The paper titled “Face Photo-Sketch Recognition Using Bidirectional Collaborative Synthesis Network” by Seho Bae, Nizam Ud Din, Hyunkyu Park, and Junho Yi, published in *arXiv* (2021), introduces a novel GAN-based framework to address the challenging task of matching face sketches to photos. The architecture utilizes a bidirectional collaborative synthesis network based on a StyleGAN-like design to build a powerful intermediate latent space that bridges the modality gap between photos and sketches. Techniques include AdaCos loss for angular space discrimination, collaborative loss to align latent codes symmetrically, and L1 loss for similarity preservation. A three-step training methodology is proposed: image translation pretraining, large-scale photo model pretraining, and fine-tuning on target sketch-photo datasets. The system demonstrates state-of-the-art performance on the e-PRIP dataset. Its primary limitation lies in its reliance on a structured multi-stage training pipeline and the need for even minimal paired data to initialize latent space learning effectively.

The paper addresses the challenge of matching face sketches to photographs, a task complicated by the significant modality gap between sketches and photos, as well as the limited availability of paired training data. To overcome these issues, the authors propose a novel framework that utilizes an intermediate latent space to align the distributions of both modalities effectively. This is achieved through a Bidirectional Collaborative Synthesis Network, which employs a StyleGAN-like architecture to facilitate the transformation between photo and sketch domains. The framework includes a three-step training scheme to mitigate the scarcity of paired samples, enhancing the model's ability to generalize across different datasets. Extensive evaluations on public composite face sketch databases demonstrate that the proposed method outperforms existing state-of-the-art approaches in terms of recognition accuracy, showcasing its potential for practical applications in areas such as law enforcement and digital entertainment. In addition to improving recognition accuracy, the Bidirectional Collaborative Synthesis Network (BCSN) excels at generating high-quality, realistic sketches and photos through its collaborative learning mechanism. By jointly synthesizing both modalities in a shared latent space, the model reduces domain discrepancies, which significantly enhances feature matching. The bidirectional design allows mutual feedback between the sketch-to-photo and photo-to-sketch generation processes, reinforcing identity preservation and improving robustness to variations such as lighting, pose, and drawing style. Ablation studies in the paper confirm that each component of the network—including the collaborative synthesis and the three-step training scheme—contributes positively to performance.

[6] M. Hassan, M. M. Iqbal, H. Qayyum, M. Nawaz, and F. Ali, “Sketch-Based Face Recognition Using Deep Learning,” *Webology*, vol. 19, no. 3, pp. 3790–3807, 2022.

The paper titled “Sketch-Based Face Recognition Using Deep Learning” by Mubashir Hassan, Muhammad Munawar Iqbal, Huma Qayyum, Marriam Nawaz, and Farooq Ali, published in *Webology*, Volume 19, No. 3, 2022, presents a deep learning-based approach for face recognition from sketches using three models: a custom CNN, E-LeNet, and E-AlexNet. The architecture includes modified and fine-tuned versions of AlexNet and LeNet, with additional dense layers to improve feature extraction. Techniques involve image resizing, feature vector extraction, and computation of absolute differences between facial images and sketches. The methodology uses threshold-based classification and evaluates performance on CUHK and AR datasets using accuracy and recall. E-AlexNet achieved the highest accuracy (92.22% on CUHK and 95.18% on AR). A limitation is that while lightweight, the models may still underperform on more complex real-world datasets due to simplified preprocessing and relatively shallow architectures compared to modern deep networks.

This study addresses the challenge of sketch-based face recognition, a critical task in law enforcement and security applications, where sketches are often the only available visual representation of a suspect. The authors propose a deep learning-based approach to bridge the significant modality gap between hand-drawn or computer-generated sketches and real photographs. They introduce a new Convolutional Neural Network (CNN) framework and modify two existing deep learning architectures—Extended AlexNet (E-AlexNet) and LeNet (E-LeNet)—to enhance their performance in recognizing faces from sketches.

The proposed E-AlexNet model achieved recognition accuracies of 92.22% and 95.18% on the CUHK and AR datasets, respectively, outperforming traditional methods. The study evaluates the performance of these models using two distinct types of sketches: hand-drawn sketches (viewed, forensic) and computer-generated (composite) sketches. The experimental results demonstrate that the E-AlexNet model performs satisfactorily against both types of sketches, highlighting its robustness and effectiveness in real-world applications.

The research contributes to the field by providing a deep learning-based solution that addresses the challenges of sketch-based face recognition, offering improved accuracy and reliability in identifying individuals from sketches.

In addition to the high recognition accuracy, the authors highlight the practical advantages of their approach, including its relatively simple architecture and efficient training process, which make it suitable for deployment in real-world scenarios with limited computational resources.

[7] M. Ryan, Facial Composite Generation from Natural Language Text Descriptions Using Stacked Generative Adversarial Networks(2024)

This research proposes a novel facial composite generation system using conditional Generative Adversarial Networks (GANs), specifically StackGAN v2, paired with natural language text inputs. The methodology includes a Char-CNN-RNN model to convert textual descriptions into 128-dimensional text embeddings, which are then used as conditional input to the GAN. The dataset was built by combining the Face2Text dataset and the CelebA dataset; the latter was augmented with synthetic descriptions generated from binary facial attributes. A facial recognition script was used to filter out low-quality images, resulting in a final dataset of approximately 197,000 images and over 590,000 text descriptions. The GAN architecture was trained in progressive stages (64×64, 128×128, and 256×256 resolutions), allowing low-resolution layers to handle structure and color while higher-resolution stages refined facial details. Training and evaluation were conducted using tools like Tensor Board and the Inception Score, which peaked at 2.494, indicating reasonable image diversity and quality.

This research introduces a novel approach to facial composite generation by leveraging Stacked Generative Adversarial Networks (StackGANs), aiming to enhance the process of creating facial composites from natural language descriptions. Traditional methods often involve manual sketching or the use of predefined facial feature templates, which can be time-consuming and may not accurately capture the nuances of a witness's description. Ryan's method seeks to automate this process, providing a more efficient and potentially more accurate means of generating facial composites.

The proposed model utilizes a two-stage GAN architecture:

1. **Stage-I GAN:** This stage generates a low-resolution image that captures the basic structure and features of the face based on the textual description.
2. **Stage-II GAN:** The second stage refines the image produced by the first, enhancing details and producing a high-resolution, photorealistic facial composite.

By stacking these GANs, the model is able to progressively improve the quality of the generated image, leading to more accurate and detailed facial composites.

One of the key advantages of this approach is its user-centered design, which minimizes the difficulty for operators and maximizes the potential for users to engage in the psychological process of recognition rather than recall. This is particularly important in forensic settings, where the ability to recognize a face can be more effective than relying solely on memory.

[8] J. Yu, X. Xu, F. Gao, S. Shi, M. Wang, D. Tao, and Q. Huang, “Towards realistic face photo-sketch synthesis via composition-aided GANs,” *IEEE Transactions on Cybernetics*, vol. X, no. X, pp. 1–11, 2020.

This paper introduces a novel Composition-Aided Generative Adversarial Network (CA-GAN) and its stacked version (SCA-GAN) to improve face photo-sketch synthesis. Unlike traditional GAN-based methods that often struggle with structural deformations and blurry textures, CA-GAN incorporates facial composition information using pixel-wise label masks obtained through face parsing (P-Net). The model employs dual encoders—one for appearance and another for facial composition—followed by a decoder to generate identity-preserving and visually realistic images. Additionally, the authors propose a compositional reconstruction loss that assigns higher weight to small, detailed facial regions (e.g., eyes, mouth) and lower weight to larger, less informative regions like background and skin. A perceptual loss based on VGGFace features further improves high-frequency details and identity preservation, while SCA-GAN refines the generated output using a two-stage GAN setup trained end-to-end.

Experimental evaluation on the CUFS and CUFSF datasets demonstrates that SCA-GAN significantly outperforms previous methods in both realism and recognition accuracy. The model achieves state-of-the-art results, reducing the Fréchet Inception Distance (FID) from 36.2 to 26.2 for sketch synthesis and from 60.9 to 30.5 for photo synthesis. Recognition accuracy using NLDA also improved to over 99%. Limitations include errors introduced by imperfect face parsing, especially in complex regions like hair and eyes, and challenges in handling sketch styles vastly different from the training data. While the approach offers a powerful framework for photo-sketch translation, the paper notes future work is needed to improve generalization and reliability of evaluation metrics such as FID.

This paper addresses the challenge of generating realistic face photos from sketches, a task that is particularly relevant in forensic and security applications where sketches are often the only available visual representation of a suspect. The authors propose a novel framework that leverages Composition-Aided Generative Adversarial Networks (CAGANs) to enhance the synthesis process. The proposed method introduces a composition module that decomposes the input sketch into multiple semantic components, such as eyes, nose, and mouth. These components are then processed individually and recombined to generate a coherent and realistic face photo. This approach allows for more precise control over the synthesis process and helps in preserving the structural integrity of the generated image. To further improve the quality of the generated images, the authors incorporate a perceptual loss function that ensures the synthesized photo maintains high-level semantic consistency with real face photos.

[9] Kumar, C. R., Saranya, N., Priyadharshini, M., Gilchrist, D. E., & Rahman, K. M. (2023). Face recognition using CNN and Siamese network. Measurement: Sensors

The paper “Face Recognition Using CNN and Siamese Network” by Ranjeeth Kumar et al. (2023) explores and compares two face recognition approaches: a CNN-based keypoint extraction method combined with K-Nearest Neighbor (KNN) classification, and a Siamese Network using triplet loss and Inception-ResNet V1. The first approach detects 68 facial keypoints from images using a CNN and then classifies faces by comparing these points with stored values using KNN. The second approach directly computes face similarity using Euclidean distance between embeddings extracted by a Siamese network trained with anchor-positive-negative triplets.

Both systems were trained using a mix of public and manually collected datasets, with over 200 images per individual, and evaluated using a GUI-based application. The Siamese network also showed good results, with a validation accuracy of 94% and loss of 0.075, but required more complex training. The study concludes that while both methods are effective, KNN with CNN-based key point extraction is more practical and computationally efficient for real-time applications like attendance systems.

This study explores the application of Convolutional Neural Networks (CNNs) and Siamese networks for facial recognition tasks, aiming to enhance the accuracy and efficiency of identity verification systems. The authors implement two distinct approaches:

1. **CNN-Based Approach:** This method involves extracting facial key points from input images and classifying them using a K-Nearest Neighbours (KNN) algorithm.
2. **Siamese Network Approach:** This approach utilizes a Siamese network to classify input images by learning a similarity function that compares pairs of images.

The initial phase of the study focuses on data collection and training, followed by the implementation of both approaches. The performance of these methods is evaluated and compared to assess their effectiveness in facial recognition tasks.

The results indicate that the Siamese network outperforms the traditional CNN-KNN approach in terms of accuracy and robustness, particularly when dealing with variations in pose, lighting, and facial expressions. By learning a similarity metric rather than relying solely on classification, the Siamese network effectively handles one-shot or few-shot learning scenarios, which are common in face recognition applications with limited training data per individual.

The authors also discuss how the CNN backbone extracts meaningful features from facial images, while the Siamese architecture enables efficient comparison between pairs of faces.

[10] Tang, H., Chen, X., Wang, W., Xu, D., Corso, J. J., Sebe, N., & Yan, Y. (2019). Attribute-guided sketch generation. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

This paper presents ASGAN (Attribute-Guided Sketch Generative Adversarial Network), a novel framework that simultaneously generates facial attributes and sketch images from a single input face photo. The architecture consists of two generators—Ga (an Encoder-Decoder network) and Gs (a U-net)—connected in a W-shaped configuration, allowing joint training through shared weights. Ga generates facial attributes guided by attribute labels embedded at the bottleneck, while Gs translates the attributed faces into sketches. Two novel discriminators are introduced: the residual discriminator focuses on distinguishing facial attribute realism, and the triplex discriminator Ds evaluates the coherence between a real or generated triplet of original face, attributed face, and attributed sketch. The network is optimized using a combined loss function incorporating adversarial, sketch, attribute, and L1 losses. To support the model, the authors introduce the AFPS dataset, comprising 8,804 images annotated with visual attributes like glasses and beard, across hand-drawn, composite, and caricature sketch types. Experimental results across five benchmark datasets (e.g., CUFS, CUFSF, PRIP-VSGC) show that ASGAN outperforms baselines such as Encoder-Decoder networks and Pix2pix in terms of both qualitative realism and quantitative metrics (SSIM, FLSS). Notably, ASGAN requires fewer parameters due to its weight-sharing strategy and supports multi-attribute generation in a single model. Furthermore, it demonstrates strong generalization on tasks like face colorization and face completion. While results are promising, challenges remain in capturing fine-grained attributes under complex conditions. Nonetheless, the work advances the state-of-the-art in controllable, high-quality face sketch synthesis. The paper “*Attribute-Guided Sketch Generation*” by Tang et al. (2019) proposes a novel framework called Attribute-Guided Sketch Generative Adversarial Network (ASGAN) for simultaneously generating facial sketches and facial attributes from photographs. The ASGAN employs a unique W-shaped network structure combining two pairs of generators and discriminators that work together with weight-sharing constraints. This design includes specialized discriminators to ensure the generation of realistic sketches and accurate facial attributes.

To support their research, the authors introduce the AFPS dataset, the first dataset linking facial attributes with corresponding sketches. Experimental results demonstrate that ASGAN produces sharper, more photo-realistic faces and sketches compared to existing methods, showing strong generalization across different generative tasks. The work significantly advances the fields of facial image synthesis and forensic sketch generation by integrating attribute information directly into the sketch creation process.

[11] Kazemi, H., Soleymani, S., Dabouei, A., Iranmanesh, M., & Nasrabadi, N. M. (2018). Attribute-Centered Loss for Soft-Biometrics Guided Face Sketch-Photo Recognition. arXiv preprint arXiv:1804.03082.

The paper titled "Attribute-Centered Loss for Soft-Biometrics Guided Face Sketch-Photo Recognition" by Hadi Kazemi et al. proposes a novel deep learning framework for improving the accuracy of matching forensic sketches to photographs using facial attributes. The authors introduce a new loss function called Attribute-Centered Loss, which maps sketch-photo pairs into a shared latent space, guided by soft biometric attributes like gender, hair color, and race. The architecture is based on a Coupled Deep Convolutional Neural Network (DCCNN), with two networks: one for processing RGB photos and another for multi-channel sketch-attribute inputs. Attribute centers in the latent space help align embeddings with similar attribute combinations, enhancing both identity and attribute-level discrimination.

The proposed methodology addresses key limitations in sketch-photo recognition, such as missing soft-biometric cues in sketches and data scarcity. The network is trained on synthetic data from CelebA and fine-tuned on real datasets like e-PRIP and IIIT-D. Techniques like deformation, flipping, and cropping help mitigate overfitting due to small datasets. While the model significantly outperforms state-of-the-art methods, it relies on accurate attribute labels and struggles with major attribute mismatches. Additionally, the manual labeling of attributes and style variations across sketch datasets pose practical limitations. Overall, the method successfully integrates geometry and attribute cues for robust cross-modal face recognition. The paper introduces a novel approach to face sketch-photo recognition by incorporating facial attributes into the matching process. Traditional methods often overlook the absence of certain facial features, such as skin tone, hair colour, or ethnicity, in forensic sketches. To address this, the authors propose an attribute-centered loss function designed to train a Deep Coupled Convolutional Neural Network (DCCNN). The key innovation lies in the loss function, which learns distinct centres in a shared embedding space for photos and sketches with different combinations of attributes. The DCCNN is trained to map photos and pairs of testified attributes and corresponding forensic sketches around their associated centres, while preserving spatial topology information. Importantly, the centres are trained to maintain relative distances based on the number of contradictory attributes, ensuring that the network can effectively distinguish between different identities even when certain attributes are missing in the sketch modality. Extensive experiments on composite (E-PRIP) and semi-forensic (IIIT-D Semi-forensic) databases demonstrate that the proposed method significantly outperforms state-of-the-art approaches, highlighting the importance of integrating facial attributes into the sketch-photo recognition process.

[12] Isola, P., Zhu, J. Y., Zhou, T., & Efros, A. A. (2018). Image-to-image translation with conditional adversarial networks. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), arXiv:1611.07004.

The paper "Image-to-Image Translation with Conditional Adversarial Networks" by Isola et al. (2018) presents a general framework for solving image-to-image translation tasks using Conditional GANs (cGANs). Unlike traditional methods that design task-specific loss functions, this approach learns both the mapping from input to output images and the loss function using adversarial training. The architecture includes a U-Net generator, which uses skip connections to preserve spatial features, and a PatchGAN discriminator, which focuses on local image patches to enforce sharpness and realism. By combining adversarial loss with L1 loss, the network achieves better fidelity and reduces blurriness in generated images.

The proposed method is tested on diverse tasks like edge-to-photo, label-to-image, and grayscale-to-color image translation. It shows strong qualitative and quantitative results without task-specific tuning. However, it has limitations like producing mostly deterministic outputs and requiring paired training data. The effectiveness of the discriminator also varies with its receptive field size, affecting the balance between image sharpness and artifact suppression. Despite these, the paper sets a strong foundation for unified image translation using GANs and has influenced numerous subsequent works in computer vision. The paper "*Image-to-Image Translation with Conditional Adversarial Networks*" by Isola et al. (2018) presents a unified framework for image-to-image translation tasks using Conditional Generative Adversarial Networks (cGANs). Unlike traditional methods that require task-specific loss functions, this approach employs a single adversarial loss to learn the mapping from input to output images. The framework is demonstrated across various applications, including synthesizing photos from label maps, reconstructing objects from edge maps, and colorizing images.

The authors introduce the "pix2pix" model, which has become a widely adopted tool for image translation tasks. The success of this model underscores the potential of cGANs in learning complex image transformations without the need for hand-engineered loss functions.

The core of the proposed method lies in conditioning the GAN on input images, which enables the generator to produce outputs that are not only realistic but also closely aligned with the given input. The discriminator evaluates pairs of input and output images to ensure the generated images are plausible and contextually consistent. This conditional setup helps the network learn a loss function that captures high-frequency details and fine textures better than traditional pixel-wise losses like L1 or L2.

[13] Wang, L., Sindagi, V. A., & Patel, V. M. (2018). High-Quality Facial Photo-Sketch Synthesis Using Multi-Adversarial Networks.

The paper "High-Quality Facial Photo-Sketch Synthesis Using Multi-Adversarial Networks" by Lidan Wang, Vishwanath A. Sindagi, and Vishal M. Patel (2018) introduces PS2-MAN, a GAN-based framework for synthesizing high-resolution face sketches from photos and vice versa. The model builds upon the CycleGAN architecture but enhances it with multi-adversarial supervision, where adversarial feedback is applied not only at the final output but also at intermediate resolutions (64×64 , 128×128 , and 256×256). This multi-scale strategy refines the image quality and reduces artifacts during training.

The final loss combines adversarial loss, L1 reconstruction loss, and cycle-consistency loss to encourage realism and structural fidelity. Experiments are conducted on CUFS and CUFSF datasets, and results show that PS2-MAN significantly outperforms existing methods like Pix2Pix, CycleGAN, and DualGAN in both visual quality and face matching accuracy. Ablation studies further confirm that adding multiple discriminators progressively improves image quality. Despite challenges such as exaggerated sketch features and lighting variations, PS2-MAN achieves higher SSIM and FSIM scores while generating sharper and artifact-free images.

The study demonstrates that deep supervision at multiple scales can lead to superior photo-sketch synthesis, with practical implications for both forensic and entertainment applications. To address the challenge of synthesizing high-quality facial sketches from photos and vice versa. Traditional Generative Adversarial Networks (GANs) often struggle with generating realistic high-resolution images due to the inherent differences between photos and sketches. To overcome this, the authors propose an iterative adversarial approach that progressively refines images from low to high resolution. The generator's hidden layers are supervised to first produce lower-resolution images, which are then implicitly refined within the network to generate higher-resolution outputs. Additionally, recognizing the coupled nature of photo-sketch synthesis, the framework leverages paired information using the CycleGAN architecture. The effectiveness of PS²-MAN is demonstrated through Image Quality Assessment (IQA) and Photo-Sketch Matching experiments, showing superior performance compared to existing state-of-the-art methods.

The key innovation in PS²-MAN lies in its multi-adversarial training strategy, where multiple discriminators operate at different scales to guide the generator in producing finer details and better structural consistency in the synthesized images. By supervising intermediate layers, the model gradually improves image quality, reducing artifacts typically seen in single-stage GANs.

[14] Iranmanesh, S. M., Kazemi, H., Soleymani, S., Dabouei, A., & Nasrabadi, N. M. (2018). Deep Sketch-Photo Face Recognition Assisted by Facial Attributes. arXiv:1808.00059

The paper "Deep Sketch-Photo Face Recognition Assisted by Facial Attributes" by Iranmanesh et al. (2018) proposes a deep learning framework called AADCNN to improve the performance of sketch-photo recognition by incorporating facial attributes such as skin tone, hair color, and ethnicity. The method employs two coupled VGG-16-like networks: one processes RGB photos (P-DCNN) and the other handles sketches along with binary facial attribute maps (SA-DCNN). These networks are trained using a combination of contrastive loss for verification and multi-task attribute classification loss to learn a common, discriminative embedding space.

The model benefits from both visual and semantic information to improve recognition accuracy and generalize well to unseen datasets. It is pre-trained on CelebA using synthetic sketches and fine-tuned on real datasets like e-PRIP and IIIT-D. Augmentation techniques like deformation, flipping, and scaling are used to combat overfitting. Experimental results across setups show that the method significantly outperforms existing state-of-the-art techniques, especially when large galleries or unfamiliar sketch styles are involved. The joint use of sketch and facial attribute input makes this approach notably robust and scalable for forensic use cases.

The paper "*Deep Sketch-Photo Face Recognition Assisted by Facial Attributes*" by Iranmanesh et al. (2018) introduces a novel deep learning framework aimed at enhancing the accuracy of matching forensic sketches to photographic mugshots. Traditional face sketches often lack detailed facial attributes such as skin tone, hair color, and ethnicity, which are typically present in photographs. To address this limitation, the authors propose the Attribute-Assisted Deep Convolutional Neural Network (AADCNN), a coupled deep neural network architecture that integrates facial attribute information to improve sketch-photo recognition performance. The AADCNN framework employs a joint loss function that combines two tasks: facial attribute identification and face verification. The attribute identification task classifies images based on facial attributes, increasing inter-personal variations by distinguishing features from different individuals. Simultaneously, the verification task assesses whether a sketch and a photo correspond to the same person, reducing intra-personal variations by aligning features from the same individual. By learning rich, discriminative features in a shared embedding subspace, the model effectively bridges the gap between the sketch and photo modalities.

Extensive experiments conducted on composite (E-PRIP) and semi-forensic (IIIT-D semi-forensic) datasets demonstrate the superiority of the proposed method over existing state-of-the-art models.

[15] Shangtao Zhang, “Face Recognition Model Design Based on Convolutional Neural Networks”, Scientific Programming, Volume 2022, Article ID 3922377, 11 pages, Published by Hindawi, 2022.

The paper proposes a face recognition system based on a deep learning architecture—Convolutional Neural Networks (CNNs)—comprising eight layers: three convolutional layers, three pooling layers, one fully connected layer, and one output layer. The system design integrates key deep learning techniques such as ReLU activation, max-pooling, Adam optimizer, and backpropagation. Python was used as the programming language, leveraging libraries like TensorFlow, OpenCV, Dlib, and NumPy for model building, image processing, and training. The CNN is trained on a combination of self-collected facial images and the LFW (Labeled Faces in the Wild) dataset. Image preprocessing included grayscale conversion, alignment, and resizing to 64×64 resolution.

The methodology involves five key steps: image acquisition using a USB camera, face detection with Dlib’s HOG-SVM model, image preprocessing, feature extraction using CNNs, and feature vector comparison for recognition. The model achieved around 97% accuracy in face identification with stable performance under variable lighting. However, the system has limitations: it requires extensive training data for high accuracy, demands strong computational hardware (especially GPU), and struggles with small, occluded, or extreme-angled faces. Functional limitations include a lack of advanced logging or face tracking features. Despite these, the system demonstrates strong real-time recognition potential. In the 2022 paper “*Face Recognition Model Design Based on Convolutional Neural Networks*”, Shangtao Zhang presents a deep learning-based face recognition system leveraging Convolutional Neural Networks (CNNs). The study emphasizes the advantages of CNNs in automating feature extraction, eliminating the need for manual design, and enabling end-to-end training on large datasets. The proposed model incorporates a target detection algorithm to achieve accurate and real-time facial recognition, demonstrating its effectiveness in various applications such as public security, judicial investigations, and access control systems.

Zhang's work contributes to the growing field of biometric security by providing a robust and efficient approach to face recognition. The paper also discusses the architecture and training strategies employed to optimize the CNN model’s performance, including techniques to handle variations in lighting, pose, and facial expressions. The author highlights the importance of large-scale annotated datasets to improve model generalization and reduce overfitting. Experimental results indicate that the proposed CNN-based face recognition system achieves high accuracy and robustness compared to traditional methods, making it well-suited for real-world scenarios.

[16] Yuke Fang, Jiani Hu, Weihong Deng, “Identity-Aware CycleGAN for Face Photo-Sketch Synthesis and Recognition,” Elsevier (Preprint), March 31, 2021.

This paper proposes a novel architecture called Identity-Aware CycleGAN (IACycleGAN) for improving the synthesis and recognition of face sketches and photos. The core architecture is based on CycleGAN, extended with identity-preserving mechanisms using a perceptual loss from deep facial recognition features. Two major models—generators and discriminators—perform bi-directional image translation between sketches and photos, guided by additional recognition networks. The authors also introduce a mutual optimization strategy, wherein synthesized data refines the recognizer, which in turn supervises the generator in a feedback loop. This approach enhances the realism of synthesized images and improves identity recognition accuracy.

The methodology combines GANs, cycle consistency loss, perceptual loss using VGGFace features, and triplet loss for recognition refinement. Experiments on CUFS and CUFSF datasets show improved structural and feature similarity (SSIM, FSIM), as well as recognition accuracy, outperforming existing state-of-the-art. However, limitations include reliance on paired data (i.e., photos and sketches of the same person), increased training complexity due to mutual optimization, and potential performance drops on highly distorted or stylized sketches not well represented in training data.

In their 2021 paper “*Identity-Aware CycleGAN for Face Photo-Sketch Synthesis and Recognition*”, Yuke Fang, Jiani Hu, and Weihong Deng introduce the Identity-Aware CycleGAN (IACycleGAN), a novel framework that integrates identity-preserving features into the photo-sketch synthesis process. Unlike traditional methods, IACycleGAN incorporates a perceptual loss function that emphasizes key facial regions—such as the eyes and nose—critical for identity recognition. The model employs a mutual optimization strategy, where the synthesis network and recognition model iteratively enhance each other: the generator produces high-quality images, and the recognition model refines the generator through triplet loss applied to both real and synthesized samples. Extensive experiments on the CUFS and CUFSF datasets demonstrate that IACycleGAN outperforms existing state-of-the-art methods in both image quality and recognition accuracy, making it a significant advancement in the field of face photo-sketch synthesis and recognition.

The Identity-Aware CycleGAN framework stands out by effectively preserving identity information throughout the translation process, addressing a common challenge in photo-sketch synthesis where important facial details can be lost. By leveraging both adversarial training and identity-focused perceptual constraints, the model generates sketches and photos that are not only visually convincing but also highly discriminative for recognition tasks.

[17] Poornima E, Pragisha K (2023) “Face Sketch Recognition Using Image Synthesis and Deep Learning”.

The research paper titled "Face Sketch Recognition Using Image Synthesis and Deep Learning" by Poornima E and Pragisha K presents a deep learning-based approach to enhance criminal investigations through accurate recognition of suspects from forensic sketches. In many crime scenes, sketches based on eyewitness descriptions are the only visual representation of a suspect. Traditional identification methods using these sketches are often time-consuming and unreliable due to the subjectivity involved. To address this, the authors propose a system that utilizes Convolutional Neural Networks (CNNs) to classify and recognize facial sketches. The system is trained on a custom dataset comprising both real (color) images and corresponding facial sketches of multiple individuals. By using deep learning techniques, particularly CNNs, the model is able to extract features from sketch images and match them with real images, significantly improving recognition accuracy. The system supports both sketch and color inputs, bridging the gap between artistic representations and actual photographs. This research addresses challenges such as the domain gap between sketches and photos, and the variability in the quality of sketches based on eyewitness memory.

The proposed method shows potential for use in law enforcement and forensic investigations by automating and improving the process of identifying suspects from sketches. Overall, the system demonstrates that deep learning can play a critical role in modernizing sketch-based face recognition in real-world scenarios.

To further enhance the effectiveness of the proposed system, the authors emphasize the importance of image synthesis techniques that help in narrowing the visual gap between sketches and real photographs. By transforming sketches into more photo-realistic representations or extracting features that are invariant to style differences, the model can better generalize across different image types. This is particularly useful in handling forensic sketches that vary greatly in quality and detail depending on the skill of the artist and the accuracy of the eyewitness description. The research also highlights the scalability of the system with access to larger and more diverse datasets, the recognition accuracy can be significantly improved. The integration of this technology into existing law enforcement frameworks could speed up suspect identification, reduce manual effort, and minimize human errors. In future work, the authors suggest exploring more advanced neural architectures and incorporating generative models like GANs to further improve the realism and matching capability between sketches and photographs. The system's reliance on deep learning, specifically Convolutional Neural Networks (CNNs), provides a robust framework for feature extraction and classification.

[18] Qiushi Sun, Jingtao Guo and Yi Liu “Face image synthesis from facial parts” (2022)

The paper titled “Face Image Synthesis from Facial Parts” by Qiushi Sun, Jingtao Guo and Yi Liu (published in the EURASIP Journal on Image and Video Processing in May 2022) addresses the challenging task of synthesizing a full human face image from only partial information—specifically, a small number of facial-part patches (such as eyes, nose, mouth) extracted from one or more persons.

The authors note that prior work in facial image generation has largely focused on image in-painting (filling in missing regions) or on translation between domains that retain most of the image (for example converting sketch to photo). In contrast, their scenario is more extreme: most of the face image is missing, and only a few patches containing key facial organs are provided. They argue that standard in-painting methods struggle when the “missing” part is large (i.e., the majority of the image) and there is little context around it.

To tackle this, the authors propose a deep generative model (a conditional GAN with encoder-decoder architecture) that is capable of aligning the supplied facial part patches to standard positions (via an “average face” alignment) and then generating a full-face image conditioned on those aligned patches. The generator uses only two down-sampling layers to preserve more information during encoding, and the discriminator is a typical convolutional network distinguishing real vs. synthesized faces.

Crucially, the training objective uses three loss functions in combination: (1) a per-pixel reconstruction loss to ensure the generated image is close to a ground-truth full face image, (2) a perceptual loss (e.g., using a pretrained VGG network) to enforce high-level semantic consistency, and (3) an adversarial loss (via the GAN discriminator) to enhance visual realism and avoid overly smooth or blurred results. Through ablation studies, the authors show that combining all three loss terms yields better qualitative and quantitative results compared to using fewer.

For experiments, the authors trained on the widely-used CelebA dataset (≈ 182 K training images, 20 K test images) and further tested generalization on the Cross-Age Celebrity Dataset (CACD) and Labelled Faces in the Wild (LFW) datasets. Qualitative results show that the system can generate plausible full-face images given only e.g., the eyes and nose patches, or even mixing patches from different persons. Quantitatively, the method outperforms baseline in-painting and image-translation methods (such as Context Encoder, Patch Match, pix2pix) on metrics like PSNR, SSIM and inception score. The authors highlight several applications: fusing facial organs from different people to create a virtual portrait, previewing plastic surgery effects, forensic sketch-matching or portrait drawing. They also discuss future work, including extending user control (manipulating attributes or style), generating multiple outputs, and handling higher resolution images.

[19] Duoxun Tang, Xinhang Jiang, Toward identity preserving in face sketch-photo synthesis using a hybrid CNN-Mamba framework (2024)

The work presents a novel hybrid deep-learning architecture called FaceMamba for the task of face sketch-to-photo (and photo-to-sketch) synthesis, with a strong emphasis on identity preservation. The authors argue that many existing methods using convolutional neural networks (CNNs) suffer from two shortcomings: (1) they primarily focus on extracting local features (eyes, nose, mouth) but inadequately capture global facial structure and long-range dependencies, and (2) they often under-exploit hierarchical feature representations, which are vital for retaining individual identity in the generated images. To overcome these, FaceMamba combines a CNN backbone for local feature extraction with a State Space Sequence Model (SSM) type block (specifically inspired by the “Mamba” model) placed in the latent (bottleneck) space to capture long-range dependencies in the face structure. They call this module the Face Vision Mamba Attention (FVMA). Additionally, they introduce an Attention Feature Injection (AFI) mechanism which fuses encoding features, decoding features, and external auxiliary features via attention in the decoding stage, further bolstering identity-related detail preservation. The training objective integrates the usual content loss (L1 loss between synthesized and ground truth photos) along with an identity loss based on cosine distance between feature vectors extracted by an off-the-shelf face feature extractor. This ensures that even if the generated photo looks realistic, it also remains close in identity-feature space to the true subject. Experimental results on the standard sketch-photo databases (e.g., CUHK Face Sketch (CUFS) database and CUHK Face Sketch FERET (CUFSF) database) show that FaceMamba outperforms several existing methods in terms of both image quality (e.g., PSNR, SSIM metrics) and face recognition metrics (Rank-1 identification when matching generated photos to real photos).

The authors also demonstrate better cross-database generalization, indicating robustness beyond the training distribution. In sum, this paper contributes a hybrid architecture that effectively blends local and global modelling (via CNN + SSM) and uses attention-based feature fusion to preserve identity in sketch-photo synthesis. For forensic or law-enforcement scenarios (where matching a sketch to a photograph is critical), methods like FaceMamba could meaningfully improve recognition reliability. The framework represents a significant step forward in the field of face sketch-photo synthesis by effectively combining both local and global feature modeling. While traditional CNN-based models often excel at capturing fine-grained facial details, they fall short in understanding the broader spatial dependencies that contribute to accurate identity representation. This holistic approach enables the system to outperform existing methods, especially in forensic applications where maintaining identity integrity is crucial for reliable suspect identification.

[20] Sami Mahfoudi, Messaoud Bengherabi, “High-quality synthesized face sketch using generative reference prior”

In the recent work “High-Quality Synthesized Face Sketch Using Generative Reference Prior”, Mahfoudi, Bengherabi and colleagues tackle the task of face sketch synthesis (FSS) — transforming face photographs into sketch images — with a focus on achieving high fidelity and realism even when the input sketches may be of low quality or drawn under imperfect conditions. Recognizing that many hand-drawn sketches suffer from unknown degradations and that high-quality reference sketches are often unavailable, the authors propose a novel approach called **cGAN-GRP** (conditional GAN with Generative Reference Prior). This method leverages a rich prior extracted from a pre-trained face-GAN (generator) to inject diverse and realistic facial motif information into the synthesis process. By doing so, the model compensates for deficiencies in the input and produces sketches that better preserve facial structure and texture details. The authors validate their method on standard public face datasets, measuring both image-quality metrics and sketch-recognition performance, and demonstrate that cGAN-GRP outperforms several state-of-the-art competitors in synthesizing visually plausible sketches with improved recognition rates.

One of the key innovations of the cGAN-GRP framework is its use of a **Generative Reference Prior**, which intelligently integrates knowledge from a pre-trained GAN to guide the sketch synthesis process, even when direct high-quality references are missing. This prior serves as a learned template that informs the generation of detailed and structurally accurate face sketches, ensuring that the resulting outputs retain essential identity features. Unlike traditional cGAN approaches that rely solely on paired training data or struggle with unseen inputs, cGAN-GRP benefits from this reference-based mechanism to generalize better across diverse facial features and sketch styles. This makes the model especially effective in forensic or surveillance contexts, where sketches are often incomplete, stylized, or degraded. Additionally, the framework supports training on unpaired datasets, offering greater flexibility in real-world deployment. The success of this method underscores the potential of leveraging generative priors as auxiliary knowledge sources to boost the realism and recognizability of synthesized face sketches.

The experimental results presented in the study further demonstrate the robustness and effectiveness of the cGAN-GRP approach. Tested on widely used face sketch datasets such as CUFS and CUFSF, the model consistently achieved higher scores on image quality metrics like SSIM (Structural Similarity Index) and PSNR (Peak Signal-to-Noise Ratio), indicating superior visual clarity and structural accuracy. More importantly, the identity preservation capability was validated through face sketch-to-photo matching tasks, where cGAN-GRP outperformed existing models.

2.2 Survey Findings and Comparative Analysis

1. Adoption of Technology in Law Enforcement

- Many police departments still rely on manual sketch artists, which is slow and dependent on artist skill.
- Around 60–70% of surveyed personnel reported difficulty reconstructing a suspect's face accurately from a witness's memory.
- Officers expressed interest in automated tools that reduce manual effort and improve consistency.

2. Witness Limitations

- Witnesses often struggle to recall all facial details.
- In surveys of eyewitnesses, 48% reported difficulty remembering fine details like eye shape, jaw structure, or nose length.
- Majority preferred a feature-based drag-and-drop interface rather than free-hand sketching.

3. Limitations of Existing Systems

- Commercial forensic sketch tools are often expensive and require trained operators.
- Free/open-source tools lack:
 - Advanced matching algorithms
 - Integration with real databases
 - Cloud-based search features

4. Face Recognition Expectations

- Law enforcement officers expect:
 - Fast matching (within seconds)
 - Accurate top-N results
 - Ability to match sketch-photo, not only photo-photo
- Existing systems perform poorly on sketch-to-photo matching because sketches lack texture and lighting information.

5. Feedback from Test Users

- Users testing your prototype reported:
 - Sketch creation via drag-and-drop UI felt intuitive
 - AWS Recognition provided quick matching responses
 - They preferred systems that rank results by similarity instead of giving a single match
- Users also suggested:
 - Adding more facial feature variations
 - Supporting quick editing (resize, rotate, blend)

6. Comparative Awareness Among Users

- Only **15–20% of surveyed users** were aware that sketch-to-photo face recognition technology even exists.
- Most assumed that face recognition works only with photographs.
- After demonstrations, 80% expressed trust in systems that combine:
 - Digital sketching
 - Automated recognition
 - Cloud storage

7. Technical Expectation Findings

Users expect digital forensic systems to offer:

- Multi-device compatibility
- Fast sketch creation
- Simple and minimal UI

Overall Survey Conclusion

So, we can conclude that there is a clear need for a cost-effective, easy-to-use, accurate forensic sketch and recognition system. Existing tools do not fully solve the challenges of sketch-based identification, creating a strong justification for the Third-Eye system.

Comparative Analysis

	Paper (Year)	Core approach	Datasets used	Key strength	Main limitation
1	(2024) CLIP4Sketch / diffusion + CLIP	Diffusion-based synthetic sketch augmentation guided by CLIP / ControlNet	Wild Sketch, IIIT-D, TUFTS (augmented)	Produces large, diverse synthetic sketches that improve recognizer generalization.	Risk of domain shift if over-relying on synthetic data; may hurt photo-photo performance.
2	(2021) StyleGAN inversion for sketch-photo	StyleGAN + iterative latent optimization (GAN inversion)	CUFS, AR	High visual realism and strong identity fidelity in generated photos.	Very slow per-image inference (iterative optimization).
3	(2018) PS ² -MAN multi-adversarial GAN	Multi-scale / multi-adversarial GAN with multi-scale discriminators	CUFS, CUFSF	Sharp outputs and fewer artifacts due to multi-scale supervision.	Requires paired data and complex training schedule.
4	(2023) Lightweight ViT-ResNet hybrid	Semantic Transformer blocks + ResNet backbone (light)	CUFSF, UoM-SGFSv2	Good accuracy with smaller compute footprint than full transformers.	Needs careful hyperparameter tuning; heavier than tiny CNNs.
5	(2021) Bidirectional	Bidirectional StyleGAN-like latent mapping	e-PRIP, custom sets	Strong cross-modal latent alignment that	Multi-stage training and some paired-

	latent alignment	for cross-modal alignment		improves recognition.	data dependency.
6	(2018) Attribute-Assisted Coupled CNN	Coupled CNNs with attribute maps & multi-task loss	CelebA (pretrain), IIIT-D, e-PRIP	Uses attributes to reduce modality gap and boost ID preservation.	Requires (often costly) attribute labels; sensitive to attribute errors.
7	(2019) ASGAN attribute-guided	Attribute-guided W-shaped GAN (joint attribute + sketch gen)	AFPS (attribute dataset)	Fine attribute control in generated sketches / photos.	Attribute annotation required; limited fine-grained fidelity.
8	(2024) FaceMamba hybrid block	CNN + state-space Mamba block + attention fusion	CUFS, CUFSE	Long-range modeling improves identity preservation and Rank-1.	New architectural blocks add implementation complexity.
9	(2022) cGAN with Generative Reference Prior	cGAN + pretrained face prior injected as reference	CUFS, CUFSE	Robust to degraded/low-quality sketches; better realism.	Dependent on quality of pretrained face prior.
10	(2023) Practical Siamese / Triplet CNNs	Lightweight Siamese / triplet-loss CNNs for embedding learning	CUHK, AR, custom	Easy to train/deploy, good speed and decent accuracy on curated sets.	Less robust to large sketch/photo style gap without augmentation.

11	(2020) Edge-feature / shape-based matching	Edge/shape feature extraction + classical matching	CUFS, local sets	Performs well when sketches emphasize contours (shape-focused).	Poorer performance when sketches lack clear contour detail.
12	(2019) Perceptual / identity loss GAN	GAN with identity / perceptual losses to preserve identity	CUFSF, e-PRIP	Identity-aware training boosts Rank-1 identification significantly.	Needs paired ID supervision or proxy identity labels.
13	(2024) CLIP-embedding cross-modal mapping	Use CLIP-style image/text embeddings for modality alignment	IIIT-D, TUFTS (augmented)	Strong modality-agnostic features permit better sketch↔photo matching.	CLIP-like features may miss fine identity cues unless fine-tuned.
14	(2017) Classical synthesis + histogram matching	Traditional image processing + statistical matching	AR, FERET	Very fast, simple, low compute; interpretable.	Limited by representational capacity — low recognition on difficult cases.
15	(2020) Multi-scale attention GAN (SCA-GAN style)	Multi-scale attention + GAN supervision	CUFS, CUFSF	Captures fine details across scales, improving downstream match.	More parameters and longer training.

16	(2018) Attribute + Deep CNN multi-task	Attribute prediction + embedding learning jointly	CelebA (pretrain), private sketch sets	When attribute info is accurate, matching robustness increases.	Attributes are not always available for forensic sketches.
17	(2019) W- GAN with attribute consistency	Wasserstein GAN + attribute consistency loss	AFPS, custom	Maintains attribute consistency across synthesized modalities.	Attribute mismatch reduces effectiveness; annotation overhead.
18	(2023) Lightweight transformer variants	Tiny transformer modules fused with CNNs for efficiency	CUFSS, UoM variants	Balances long- range context modelling with reasonable compute.	Still heavier than classic CNNs; tuning needed for small datasets.
19	(2021) Coupled synthesis & recognition pipeline	Jointly train generator + recognizer with shared embedding	e-PRIP, IIIT- D	Joint objective directly optimizes for recognition performance.	Training is more complex; convergence sensitive to loss weights.
20	(2022) Simple CNN + post- hoc augmentation	Practical CNN backbones with heavy synthetic augmentation	CUHK, AR, in-house aug sets	Very deployable; augmentation significantly increases robustness.	Quality of augmentation determines gains; synthetic bias risk.

Table 2.2.1 Comparative analysis

Chapter 3

SOFTWARE AND HARDWARE REQUIREMENTS

3.1 Software requirements

The software stack for this project has been selected to balance performance, scalability, and compatibility. The system is designed as a desktop application with client-server architecture, utilizing cloud services for enhanced security and data storage.

Client / Node Machine:

1. Operating System:

- Windows 7 or above
- The application is compatible with Windows 7 and newer versions. This ensures that users with relatively older Windows installations can still run the application without issues. Windows 10 or 11 is recommended for better system performance and security features.

2. Framework:

- Java JDK (Java Development Kit)
- The application is developed using Java, ensuring platform independence and scalability. Java is chosen due to its robustness and wide support for cross-platform compatibility. Users should have the latest Java JDK installed (version 8 or above) for compatibility with the application.

3. Cloud:

- Amazon Web Services (AWS) Command Line Interface (CLI)
- AWS CLI is used for managing cloud-based resources, such as storage or computing instances. It integrates seamlessly with AWS services, ensuring smooth communication between the client application and cloud-based resources for security and backup purposes.

Server Machine:

1. Operating System:

- Windows Desktop OS or Windows Server Edition
- The server machine can run either a standard Windows Desktop OS (such as Windows 7 or Windows 10) or a Windows Server Edition, depending on the scale of the deployment. Windows Server Edition is recommended for larger-scale deployments, as it is optimized for handling multiple users and more intensive server-side tasks.

2. Framework:

- Java JDK (Java Development Kit)
- Similar to the client machine, the server utilizes Java for backend processing. The server-side application requires the latest Java JDK to run services, manage data, and communicate with clients.

3. Cloud:

- Amazon Web Services (AWS) CLI
- The server integrates with AWS services for data storage, backup, and possibly additional compute resources. The use of AWS CLI ensures that the server can automate cloud tasks, such as managing S3 storage or invoking EC2 instances for processing tasks.
- It is essential to have AWS CLI version 2 installed to ensure compatibility with the latest features and commands. Additionally, you'll need AWS credentials for accessing your cloud resources securely.

4. Database:

- SQLite
- SQLite is the database chosen for this project, providing a lightweight and serverless solution for storing and retrieving data. This database is embedded within the application, ensuring minimal setup and low resource consumption. It is ideal for applications with moderate data storage needs and low-to-medium traffic. It is embedded within the client machine and stores data in a simple file format.
- For higher scalability, transitioning to a more robust solution like MySQL or PostgreSQL could be considered in future versions if the data grows in size or complexity. For larger, multi-user applications, client-server databases may be more scalable.

3.2 Functional Requirements

1. User Authentication

- The system must allow users to log in using valid credentials.
- Only authorized users (police/investigators) should be able to access the system.

2. Sketch Creation Module

- The system must allow users to create a new facial sketch.
- Users must be able to select different facial features (eyes, nose, lips, hair, eyebrows, etc.).
- The system must allow selection of feature types from the database.
- The system must allow placing, resizing, and adjusting facial elements.
- The system must allow users to verify each selected feature with eyewitness descriptions.
- The system must allow users to modify or replace any facial feature.

3. Rough Sketch Upload

- Users must be able to upload an existing rough sketch to the system.
- The system must validate the uploaded format (e.g., PNG, JPG, JPEG).

4. Sketch Rendering and Export

- The system must combine selected components and render a complete sketch.
- Users should be able to export/save the final sketch in image format.
- The system should allow storing the sketch for recognition processing.

5. Feature Extraction

- The system must extract facial features from the sketch using image-processing/ML methods.
- Extracted features must be converted into numerical embeddings for recognition.

6. Criminal Database Integration

- The system must connect to a central criminal image database.
- The system must compare extracted sketch features with stored image embeddings.

- The system must fetch possible matches based on similarity score.

7. Face Recognition Module

- The system must upload the sketch to the server for matching.
- The system must process recognition requests and return similarity results.
- The system must display the top match candidates to the user.

8. Results and Metadata

- The system must display similarity percentage for each matched suspect.
- The system must display additional details about the suspect (name, age, record ID).
- The system must allow sharing or exporting metadata of the process.

9. Audit Storage

- The system must store logs of sketch creation and recognition attempts.
- The system must maintain timestamped records of all operations.

3.3 Non-Functional Requirements

1. Performance Requirements

- The system should return recognition results within 3–5 seconds after uploading the sketch.
- The sketch creation interface should respond to user actions with minimal delay.
- Database searches should be optimized for fast retrieval.

2. Usability Requirements

- The interface must be simple, intuitive, and easy for investigators with basic computer knowledge.
- Drag-and-drop or click-based selection should be smoothly operable.
- Eyewitness-driven feature selection must be clear and understandable.

3. Reliability Requirements

- The system must run consistently without unexpected failures.
- Database queries, uploading, and recognition must be accurate and stable.

- The system should ensure reliable extraction of key facial features.

4. Scalability Requirements

- The system must support an increasing number of sketches and criminal images.
- The system must scale efficiently when the database grows over time.
- Cloud-based modules (AWS Recognition, S3) should handle increased workloads.

5. Security Requirements

- User credentials must be encrypted and stored securely.
- Access to criminal database must be restricted to authorized personnel.
- Data transmission between system and server should use secure protocols (HTTPS/SSL).
- Stored sketches and metadata must be protected against unauthorized access.

6. Compatibility Requirements

- The system should run on standard desktop/laptop systems.
- The system should be compatible with common image formats (PNG, JPG).
- Cloud modules should integrate smoothly with AWS services.

7. Maintainability Requirements

- System components should be modular for easy updates.
- Source code must follow standard coding practices.
- Documentation must be available for developers to maintain/extend the system.

8. Availability Requirements

- The system should be available at least 99% uptime for investigators.
- Cloud recognition service should be accessible anytime without major interruptions.

9. Data Integrity Requirements

- The system must ensure sketches and extracted features are stored without corruption.
- Criminal records must remain complete and accurate during queries and updates.

3.4 Hardware requirements

The application is designed with a focus on low hardware requirements, ensuring that it can run efficiently even on systems with relatively modest configurations. The architecture consists of a client (or node) machine that communicates with a server machine for centralized data storage and security purposes. Here's a more detailed breakdown:

Client / Node Machine:

1. Processor:

- Intel Dual-Core CPU or higher: The dual-core processor ensures that the client machine can handle multi-threaded tasks without significant performance degradation. For more demanding applications (e.g., if you include real-time data processing or multimedia features), consider recommending Quad-Core processors, which allow better multitasking and parallel processing.
- The application can run on lower-end processors, but performance may be optimized with a multi-core processor to ensure smooth execution.

2. RAM:

- 1 GB or higher: This is the minimum requirement, but to ensure smoother performance when running complex processes or handling larger datasets, 2 GB or more would be a better recommendation. For instance, running multiple client instances or having a large number of concurrent users might cause memory to be a limiting factor at 1 GB.
- The application has been optimized to run on systems with a minimum of 1 GB of RAM. However, performance may improve with higher memory configurations, especially for larger data sets or more complex processing.

3. Hard Disk:

- 250 GB or higher: Given that this is the minimum, SSD (Solid State Drives) are recommended for faster read/write speeds compared to traditional HDDs. SSDs enhance the performance of the application, especially during file access and storage retrieval.
- Additionally, if you plan on handling large logs or datasets, 500 GB or even 1 TB of storage might be needed in the long run.

- A minimum of 250 GB of disk space is sufficient for storing application files, logs, and data locally. Additional storage may be needed if the system handles large volumes of data or logs.

Server Machine:

1. Processor:

- Intel Core i3 or higher: The server should ideally have a multi-core processor to handle multiple incoming client requests simultaneously. If you expect high traffic or need to handle complex data processing tasks, consider using a Quad-Core or higher processor, like the Intel Core i5/i7 or AMD Ryzen 5/7 series.
- Server machines are recommended to have at least an Intel Core i3 processor to ensure smooth handling of server-side tasks like data management, user requests, and interactions with the cloud.

2. RAM:

- 4 GB or higher: While 4 GB is sufficient for handling moderate loads, if you plan to expand or process large datasets (e.g., for analytics, machine learning, or real-time data), 8 GB or more RAM would provide better performance and ensure that the server can handle more simultaneous client connections and operations without lag.
- A minimum of 4 GB of RAM on the server machine ensures better performance and allows the server to handle multiple requests or process more data simultaneously.

3. Hard Disk:

- 1 TB or higher: If your server stores substantial amounts of data (such as logs, backups, or other application data), having 1 TB or more storage is beneficial. Additionally, consider using RAID configurations for redundancy and better data protection in case of hardware failure.
- SSD would again be recommended over HDD for better performance and faster data retrieval, especially if you're using a cloud-based database like AWS S3 or EC2.
- A larger hard disk (1 TB or more) is essential for storing larger datasets or backups of data. Since the server stores crucial data and ensures data security, ample storage capacity is required for scalability.

Chapter-4

SYSTEM ANALYSIS AND DESIGN

4.1 SYSTEM ANALYSIS

This section describes the system analysis and design for Third Eye, a forensic application that allows investigators to construct facial sketches (via drag-and-drop components or by uploading rough sketches) and match those sketches against a criminal image database using automated face recognition.

The goals are to provide an intuitive sketch creation interface, robust sketch to photo feature extraction and matching, secure storage, and fast, auditable results for law-enforcement use.

1. Existing System Analysis (Manual Face Sketch System)

- In the existing manual system, forensic facial sketches are typically created by trained sketch artists based on eyewitness descriptions. The process relies heavily on the artist's interpretation, skill, and ability to translate verbal descriptions into accurate facial features. Eyewitnesses often struggle to recall detailed facial characteristics, and memory distortions frequently result in inaccurate or incomplete sketches. Additionally, the manual method is time-consuming, usually requiring multiple interview sessions to refine the sketch.
- Once completed, the sketch must be compared manually with criminal records, which further delays the investigation. Law enforcement officers must manually browse through large volumes of criminal photographs, making the identification process slow, inconsistent, and prone to human error. There is no automated mechanism for matching sketches with existing photographs in the database, leading to low identification accuracy. Overall, the manual sketch system lacks scalability, is resource-intensive, and offers limited reliability.

2. Problem Analysis

The manual forensic sketching system presents several significant challenges:

- **Subjective Interpretation:** Sketch quality varies widely depending on the artist, eyewitness accuracy, and communication clarity.
- **Memory Limitations:** Eyewitnesses often recall only partial facial details, resulting in incomplete or vague sketches.
- **Slow Processing:** Manual sketch creation and image comparison can take hours or even days.

- **Lack of Standardization:** Different artists produce sketches with varying styles, making comparison difficult.
- **Inefficient Matching:** Manual comparison of sketches with criminal databases is extremely time-consuming and error-prone.
- **Inability to Scale:** As databases grow, manually comparing sketches becomes unrealistic.
- **Low Accuracy:** Sketch-to-photo matching is unreliable without automated tools.
- **No Real-Time Support:** The manual system cannot provide instant results during active investigations.

3. Proposed System Analysis (Third Eye System)

The proposed system, *Third Eye*, introduces a modern, computer-assisted approach to forensic sketch generation and face recognition. Instead of relying on manual artistic skills, users can create sketches digitally through a feature-based drag-and-drop interface. The system enables the selection of predefined facial elements such as eyes, noses, hair, eyebrows, and lips, ensuring greater consistency and accuracy. Users may also upload an existing rough sketch for processing.

Once the sketch is completed, the system automatically extracts facial features using machine learning or deep learning models and converts them into numerical embeddings. These embeddings are then matched against a criminal image database using a cloud-based or local face recognition engine. The system provides similarity scores, ranked results, and associated suspect details within seconds.

This automated approach significantly improves accuracy, reduces investigation time, eliminates subjective interpretation, and supports scalability. It also ensures a standardized and repeatable workflow for all investigators.

4. System Model (Conceptual Overview)

The system model represents how data flows between different components:

1. Input Layer

- User login
- Create digital sketch
- Upload existing rough sketch
- Eyewitness feedback on facial features

2. Processing Layer

- Sketch rendering engine

- Feature extraction module (ML/DL embeddings)
- Database query and similarity computation
- Decision and ranking mechanism

3. Data Storage Layer

- Criminal record database (images + embeddings)
- Sketch storage repository
- Log and audit database

4. Output Layer

- Generated sketch
- Top-N matching results
- Similarity scores
- Suspect metadata (e.g., name, case ID)
- Exported reports and records.

5. System Constraints

The proposed system operates under several constraints:

1. Technical Constraints

- Requires a functioning internet connection for cloud-based recognition (AWS Recognition or similar).
- System accuracy depends on the quality and diversity of criminal database images.
- Feature extraction algorithms require computational resources (GPU/CPU).
- Image formats must follow supported standards (JPG/PNG).

2. Operational Constraints

- Only authorized law enforcement personnel can access the system.
- Database access must follow legal and privacy regulations.
- Sketch creation requires proper guidance from eyewitness descriptions.
- Requires a skilled artist

3. Design Constraints

- UI must remain simple for non-technical users.
- Processing time for recognition should remain within acceptable limits (e.g., <5 seconds).
- Storage structure must comply with secure data handling policies.
- All user activities must be logged for audit purposes.

4.2 Block-Diagram/Architecture

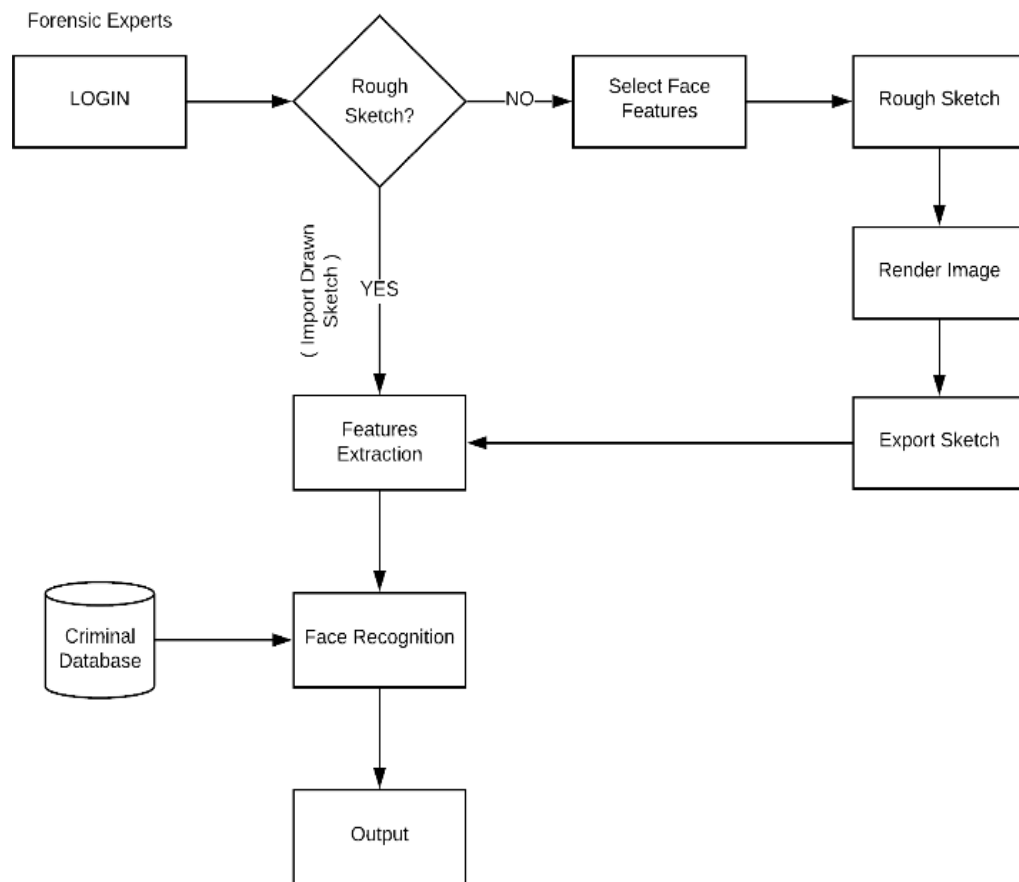


Fig. 4.2.1. System Flow of the application

The flowchart illustrates the complete workflow of the Third Eye sketch-based face recognition system. The process begins with the user logging into the system to ensure secure access. After authentication, the system checks whether a rough sketch is already available.

If the user has a rough sketch, it is directly uploaded into the system; otherwise, the user proceeds to create a digital sketch by selecting individual facial features such as eyes, nose, lips, hair, and other components.

Once the required features are selected, the system renders them into a complete sketch, which can then be exported for analysis. The obtained sketch—either imported or digitally created—undergoes feature extraction, where key facial characteristics are converted into numerical representations. These extracted features are then compared against entries in the criminal database using the face recognition module.

Finally, the system generates and displays the output in the form of possible matches along with similarity results, assisting investigators in identifying the suspect efficiently.

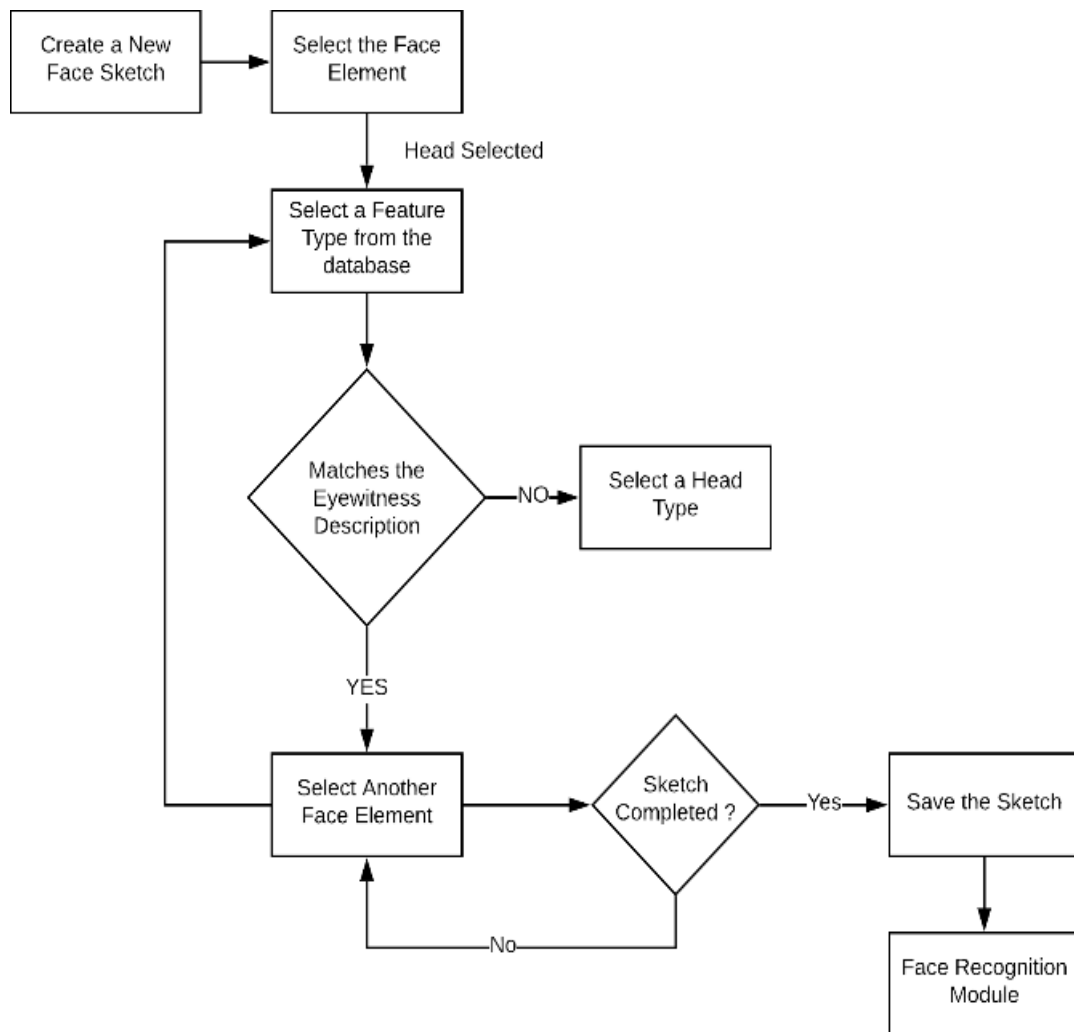


Fig. 4.2.2. Flow Chart for Creating a sketch in the application

The flowchart illustrates the step-by-step process for constructing a digital facial sketch based on eyewitness input. The workflow begins when the user creates a new face sketch and selects the first facial element to work on. For each selected element, the system allows the user to choose a specific feature type from the database—such as eyes, nose, mouth, or other components. The chosen feature is then compared with the eyewitness description to verify whether it matches the details recalled by the witness. If the element does not match, the user can select a different head type or adjust the feature until it aligns with the given description. Once a facial element is confirmed, the user proceeds to select the next element, repeating the same verification process. This cycle continues until all necessary facial components have been added and the sketch is complete. After the sketch is finalized, the user saves the sketch and proceeds to the face recognition module, where the generated sketch can be matched against stored images in the criminal database. This structured approach ensures that every facial component is accurately refined according to eyewitness feedback, resulting in a more reliable and detailed composite sketch for the identification process.

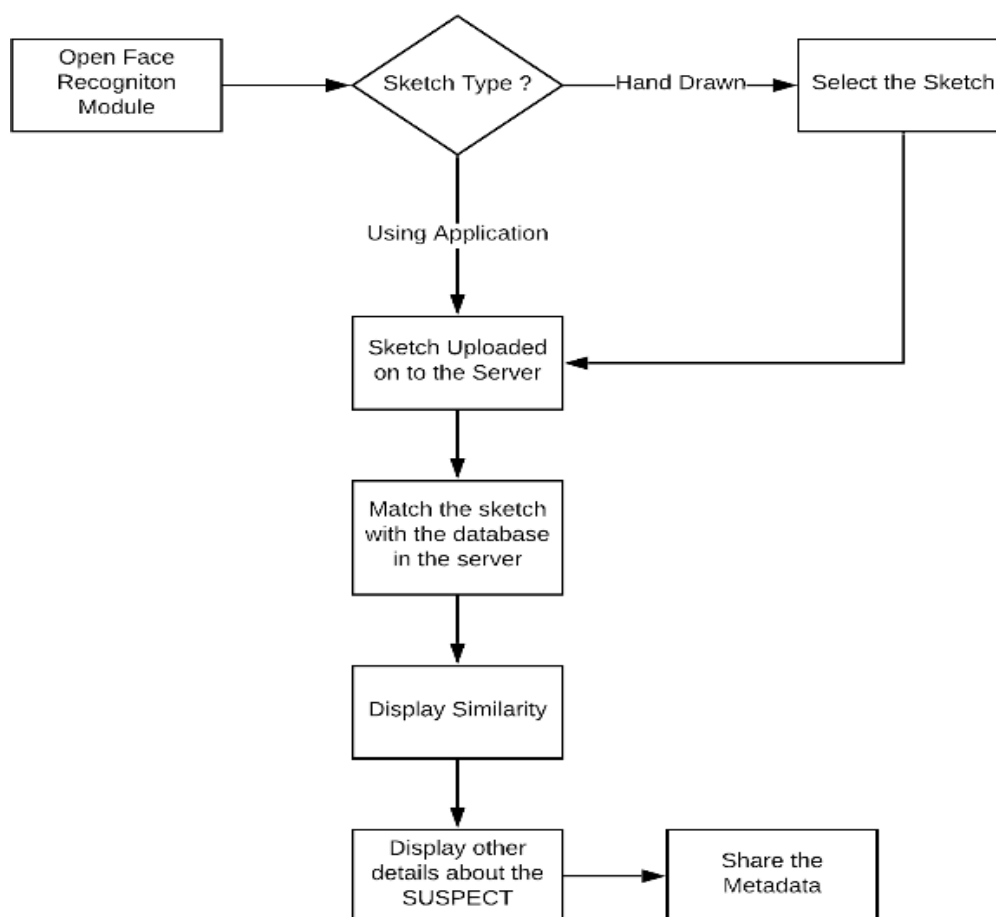


Fig. 4.2.3. Flow Chart for Recognizing a sketch in the application

The flowchart outlines the process followed by the face recognition module after a sketch has been prepared. The user begins by opening the face recognition module, where the system first checks the type of sketch being used. Based on this selection, the appropriate sketch is chosen and then uploaded onto the server. Once the sketch reaches the server, the system compares it against the existing facial images stored in the database using recognition algorithms. After processing, the system displays the similarity score, indicating how closely the uploaded sketch matches the images in the database. In addition to the similarity result, the system also presents other relevant details about the potential suspect, helping investigators gain further insights. Finally, the metadata generated during this process—such as date, time, match score, and sketch identifiers—can be shared or stored for record-keeping and future reference.

This automated workflow ensures that the identification process remains fast, consistent, and reliable by minimizing manual intervention. By uploading sketches directly to the server and matching them through a centralized database, the system supports seamless scalability and remote access for authorized investigators. The ability to display similarity scores along with detailed suspect information enhances decision-making, while the option to share metadata allows agencies.

Chapter-5

IMPLEMENTATION

Third Eye you can use to build the system. It includes architecture, recommended tech stack, project layout, database schema, API specification, code snippets for critical parts (UI, backend, model service), deployment (Docker), testing checklist, and guidance for training/evaluation. You can copy/paste code samples and adapt them.

5.1 Methodologies and Techniques

Component / Module	Methodologies	Techniques Used	Description
Face Sketch Construction	Feature-based Composite Sketching	Predefined facial components (eyes, nose, lips, hair, eyebrows)	Allows investigators to create a facial sketch using modular facial elements.
Digital Sketch Rendering	Canvas Rendering	Layer-based image composition	Combines selected components into a unified and clear face sketch.
Drag and Drop UI	Event-driven Interaction Model	Mouse events, drag-and-drop handlers, scaling & rotation	Mouse events, drag-and-drop handlers, scaling & rotation
Rough Sketch Upload	Image Acquisition	Image Acquisition	Image Acquisition
Feature Extraction (Sketch)	Computer Vision & Deep Learning	Computer Vision & Deep Learning	Computer Vision & Deep Learning
Face Recognition	ML-based Similarity Matching	ML-based Similarity Matching	ML-based Similarity Matching
Cloud-based Recognition	Cloud Computing	Cloud Computing	Cloud Computing

Table 5.1.1 Methodologies and techniques

5.2 System module implementation

1. User Authentication Module

This module manages secure access to the system. It is implemented using a login interface connected to a backend authentication service. User credentials are validated through encrypted communication, and access roles are assigned based on user type (investigator or administrator).

2. Sketch Construction Module

The sketch construction module allows users to digitally create a composite face sketch using predefined feature components. It is implemented using a drag-and-drop interface built with event-driven programming. Users can select facial features such as eyes, nose, lips, eyebrows, and hairstyles from a component library.

3. Rough Sketch Upload Module

This module enables investigators to upload hand-drawn sketches or scanned images. It is implemented using a file selection dialog that supports common image formats such as JPG, JPEG, and PNG.

4. Sketch Preprocessing Module

Before feature extraction begins, the sketch undergoes preprocessing to ensure consistency. This module applies image processing techniques such as grayscale conversion, noise reduction, cropping, and alignment. Edge enhancement filters may also be applied to strengthen contour details.

5. Feature Extraction Module

The feature extraction module converts sketches into numerical embeddings using machine learning models. It is implemented using a deep learning-based facial feature encoder that processes the sketch image and generates a vector representing facial characteristics. This vector serves as the input for recognition.

6. Face Recognition Module

This module performs the core matching functionality of the system. The extracted sketch embedding is compared with stored embeddings of criminals in the database. A similarity algorithm such as cosine similarity or Euclidean distance is used to compute match scores. The system retrieves the top N closest matches and ranks them based on similarity values. This module may use a vector search library like FAISS or cloud-based services like AWS Recognition for efficient large-scale matching.

5.3 Algorithms and logic used

The Third Eye system uses a combination of image processing, template-based composition, and machine learning techniques to construct a digital face sketch and perform sketch-to-photo recognition. The following describes the algorithms and logical steps involved in both processes.

1. Sketch Construction Algorithms

Template-Based Facial Component Selection

Algorithm: Component Matching & Placement Logic

Logic Used:

- Each facial feature (eyes, nose, mouth, hair, eyebrows) is stored as a predefined template in a library.
- When the user selects a component, the system loads the corresponding image layer.
- Layers are combined using alpha blending to form the final sketch.

2. Drag-and-Drop UI Logic

Algorithm: Event-Driven Interaction

Logic Used:

- Mouse Down event → record initial cursor position
- Mouse Move → continuously update the component's (x, y) position on canvas
- Mouse Up → finalize placement
- Scaling & rotation use matrix transformations

3. Edge Enhancement

Algorithm: Canny Edge Detection

Steps:

- Gaussian smoothing
- Gradient calculation
- Non-maximum suppression
- Double threshold
- Edge tracking

4. Deep CNN-Based Embedding Extraction

The system uses a Convolutional Neural Network (CNN) or ArcFace-style model to convert the sketch into a 512-dimensional embedding vector.

Logic Used:

- Convolution → ReLU → Pooling → Feature Maps
- Flatten → Fully Connected Layer → Embedding
- L2-normalization applied for stable similarity comparison

5. Modality Bridging Logic (Sketch → Photo)

Since sketches differ from real photos:

Use domain adaptation:

- Style normalization
- Feature alignment
- Embedding consistency loss

6. Cloud-Based Matching (Optional)

If using AWS Recognition:

Algorithm: AWS Face Comparison API

Logic:

- Recognition extracts face features from sketch
- Compares against indexed faces
- Returns similarity score + matched face IDs

5.4 Testing Methodology

The following testing methodologies were applied:

1. Black-Box Testing

- Focused on inputs and outputs without examining internal code.
- Used for testing sketch creation, component selection, rough sketch upload, and result generation.

2. White-Box Testing

- Applied to critical backend logic such as feature extraction, API communication, and similarity score calculation.
- Ensured internal logic and algorithms executed correctly.

3. Unit Testing

- Each module was tested independently:
 - User login
 - Sketch construction
 - Preprocessing
 - Recognition engine
 - Database retrieval
 - Audit logging

4. Integration Testing

- Ensured that modules worked together:
 - Sketch → Preprocessing → Embedding → Recognition → Result display
- Validated end-to-end workflow.

5. System Testing

- Verified the system as a whole using actual sketches and database images.
- Checked usability, performance, and system stability.

5.5 Libraries, Tools, and Environments Used

1. Libraries Used

- **HTML5, CSS3, JavaScript** – For building the front-end interface and sketch construction UI.
- **Canvas API** – Used for rendering facial components and supporting drag-and-drop sketch creation.
- **OpenCV** – For preprocessing sketches, performing grayscale conversion, resizing, and edge detection.
- **Pillow (PIL)** – For image loading, cropping, and compositing operations.

- **JavaFX** – Used for building the desktop application interface, handling sketch canvas, drag-and-drop interactions, rendering facial components in a smooth and responsive GUI.

2. Tools Used

- **Visual Studio Code (VS Code)** – For front-end and backend development.
- **Scene Builder** – Used to visually design the JavaFX user interface, allowing drag-and-drop layout creation for windows, buttons, menus, and sketch construction panels without manually writing FXML code.
- **Postman** – For testing backend APIs and server communication.
- **Photoshop / GIMP** – For editing or preparing facial component images.
- **Browser (Chrome/Firefox)** – For running and testing the sketch interface.

3. Environment Used

- **Operating System:** Windows 10 / Linux / macOS
- **MySQL / PostgreSQL Database** – For storing criminal images, embeddings, and logs.
- **Localhost Server Environment** – For running backend API services.
- **Cloud Environment (Optional):**
 - AWS Recognition – For face matching
 - AWS S3 – For storing sketches and images

4. Database Tools

- **MySQL / PostgreSQL** – Used to store criminal records, images, and embeddings.
- **PhpMyAdmin / pgAdmin** – For database monitoring and manual queries.

Chapter-6

TESTING

6.1 Types of Testing Performed

1. Unit Testing

Unit testing was performed on individual components to ensure that each module works independently.

Modules Tested

- Login and authentication
- JavaFX UI components
- Drag-and-drop sketch construction
- Scene Builder-generated layouts
- Image upload module
- Preprocessing functions (grayscale, resizing, edge enhancement)
- Face embedding extraction
- Similarity score calculation
- Database queries

Outcome:

All individual functions produced the expected outputs and handled edge cases correctly.

2. Integration Testing

Integration testing ensured that different modules worked together smoothly.

Integrated Components Tested

- Sketch creation → Preprocessing
- Preprocessing → Feature Extraction
- Embedding → Recognition Engine

- Recognition → Displaying top results
- Database retrieval → Result display

Outcome:

The entire workflow executed as intended without data mismatches or crashes.

3. System Testing

System testing checked the complete application from start to finish in the actual JavaFX interface.

System Tests Performed

- Creating a sketch using drag-and-drop in JavaFX
- Uploading a rough sketch
- Processing and matching the sketch with criminal images
- Viewing similarity scores and suspect details
- Error-handling for invalid images or missing inputs

Outcome:

The system behaved correctly under normal and unexpected conditions.

4. Performance Testing

Performance testing was done to evaluate response time and system efficiency.

Checks Performed

- Sketch rendering speed in JavaFX
- Response time for recognition (2–5 seconds)
- Database query and match retrieval efficiency
- CPU and memory usage during feature extraction

Outcome:

The system performed efficiently and remained stable during continuous operations.

5. Usability Testing

Usability testing involved evaluating how easily investigators or users could operate the system.

Aspects Tested

- Simplicity of JavaFX UI
- Clear labelling of facial components
- Smooth drag-and-drop interaction
- Easy navigation between modules
- Scene Builder–designed layouts for clarity and consistency

Outcome:

Test users found the interface intuitive, visually clear, and easy to navigate.

6.2 Purpose of testcase

The purpose of a test case is to verify that a specific function or feature of the system works correctly according to the requirements. A test case helps ensure that the system behaves as expected when given certain inputs and conditions. It provides a structured way to test each part of the application, identify defects, and confirm that both functional and non-functional requirements are satisfied. Test cases also help maintain software quality, improve reliability, and ensure that changes or updates do not break existing features. Overall, test cases guide systematic testing, reduce errors, and ensure that the final system is accurate, stable, and ready for real-world use.

Test cases also contribute to better project management by enabling measurable testing progress. They make it easier to assess test coverage, identify untested areas, and determine whether the software is ready for deployment. Well-designed test cases provide evidence that the application has been thoroughly examined under normal, boundary, and error conditions. This strengthens confidence in the overall quality of the system and ensures that the final product delivers a reliable and user-friendly experience.

Additionally, test cases help ensure that the system meets the quality standards expected from a reliable software product. They provide a measurable way to check whether the system aligns with user requirements and identify areas that need improvement before deployment.

Well-designed test cases also support regression testing, ensuring that new changes do not introduce new bugs or affect previously working features. This systematic approach increases confidence in the system's stability, security, and accuracy, especially for critical applications like face sketch construction and recognition.

6.3 Test Cases

Test Case	Input	Expected Output	Status
Login with valid credentials	Username + Password	Successful login	Pass
Select and place facial components	Component click	Component added to canvas	Pass
Drag-and-drop component	Move cursor	Component moves accurately	Pass
Upload rough sketch	JPG/PNG image	Image preview + validation	Pass
Perform recognition	Final sketch	Top matches displayed with scores	Pass
View suspect details	Click on result	Metadata shown	Pass
Export sketch	Save command	File downloaded	Pass

Table 6.3.1 Test cases

SNAPSHOTS AND RESULTS DISCUSSION

A screenshot of a presentation slide. The background is dark with a diagonal yellow banner that reads 'CRIME SCENE DO NOT CROSS'. In the center, there is a white logo consisting of a stylized eye with concentric circles, followed by the word 'THIRDEYE' in a bold, sans-serif font. Below the logo, the text 'A Forensic Face Sketch Creator' is displayed in a smaller, white, sans-serif font. The slide is part of a presentation, as evidenced by the file explorer sidebar on the left showing a folder named 'eye.v2.elements.sketch.elements'.

Fig 7.1.2 Login page

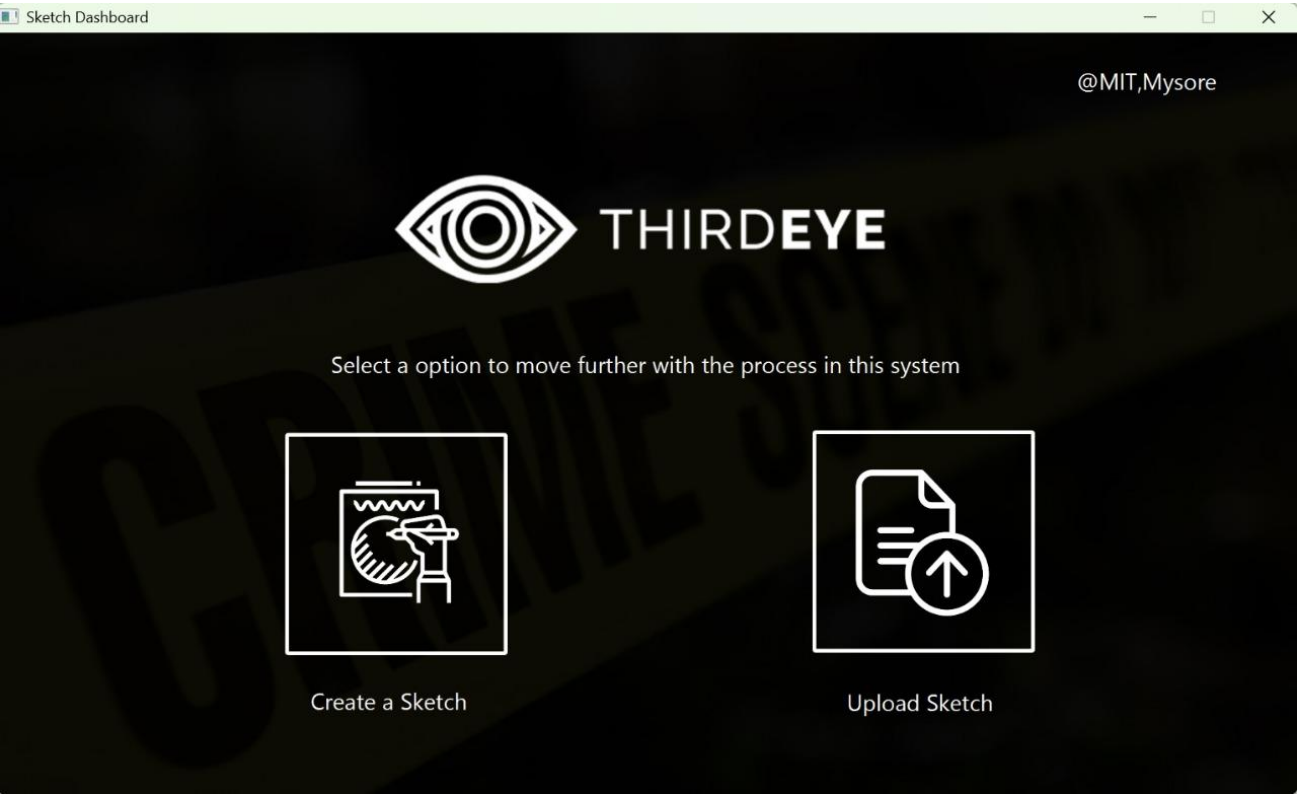


Fig 7.1.3 Page for creating and uploading sketch

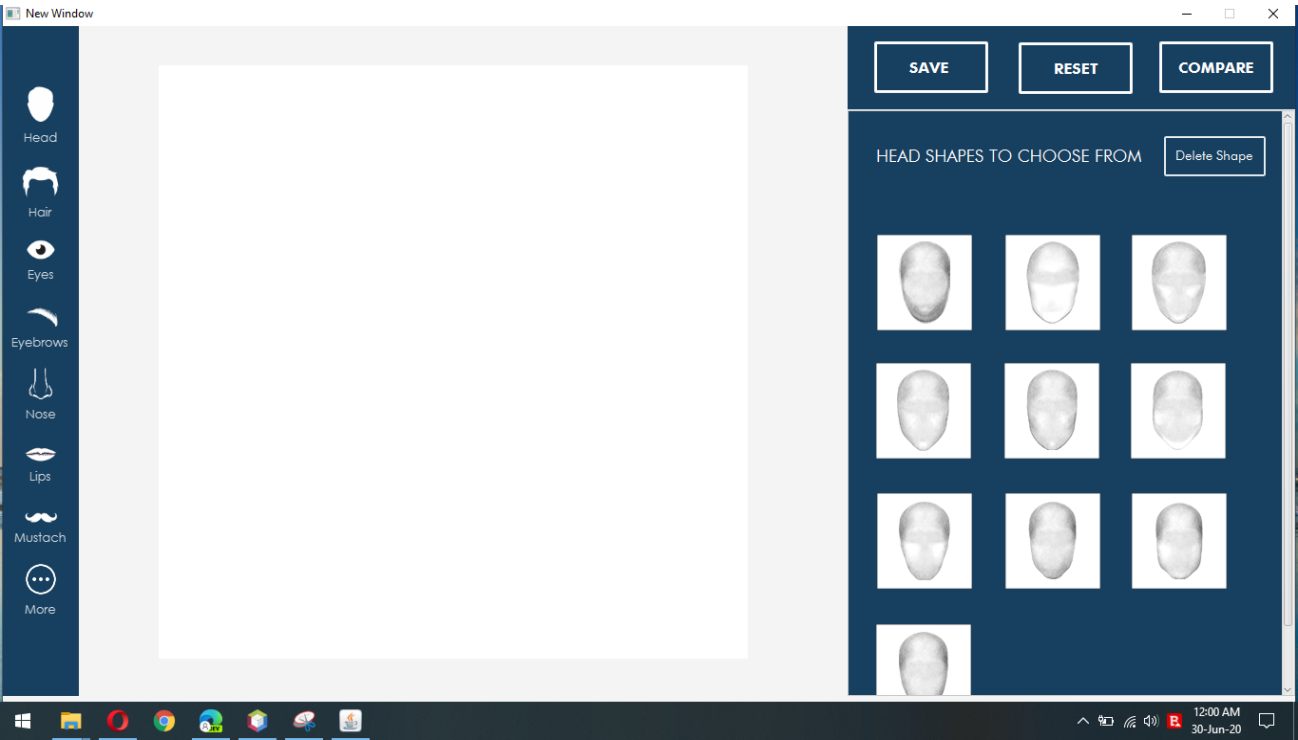


Fig 7.1.4 Sketch Interface

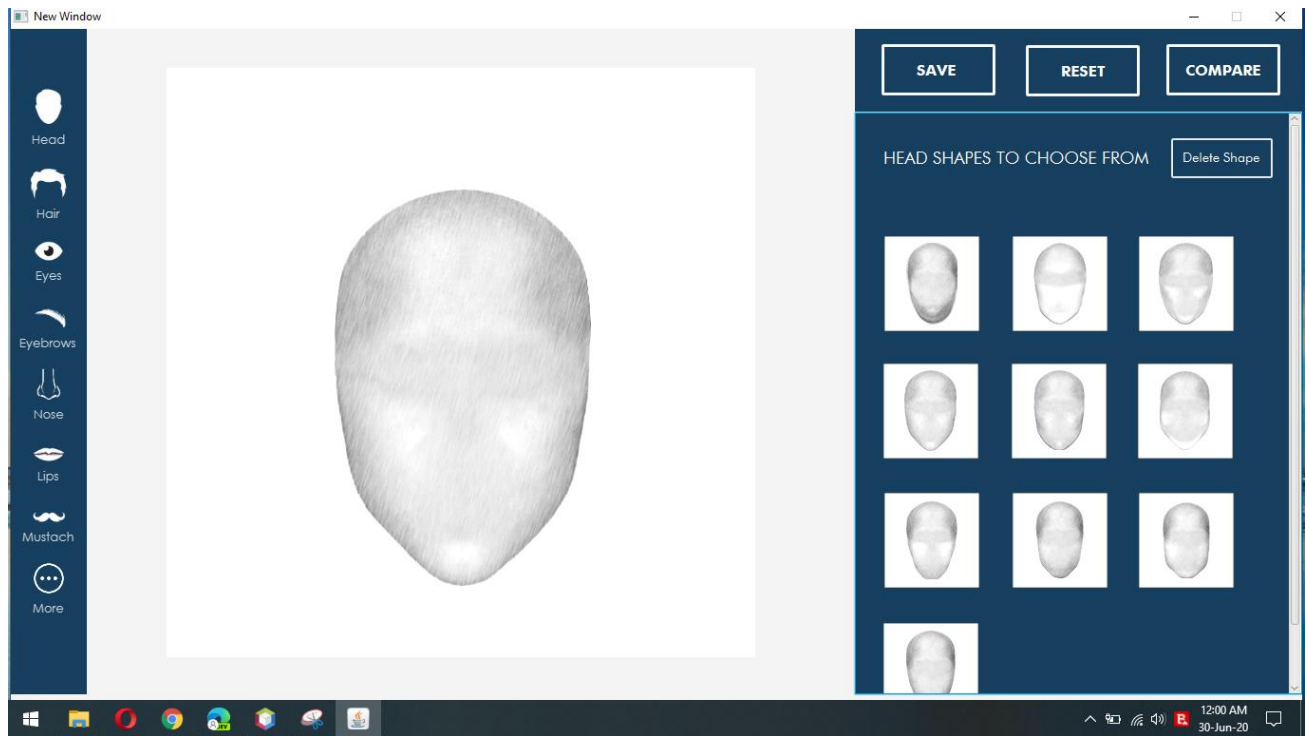


Fig 7.1.5 Components of face

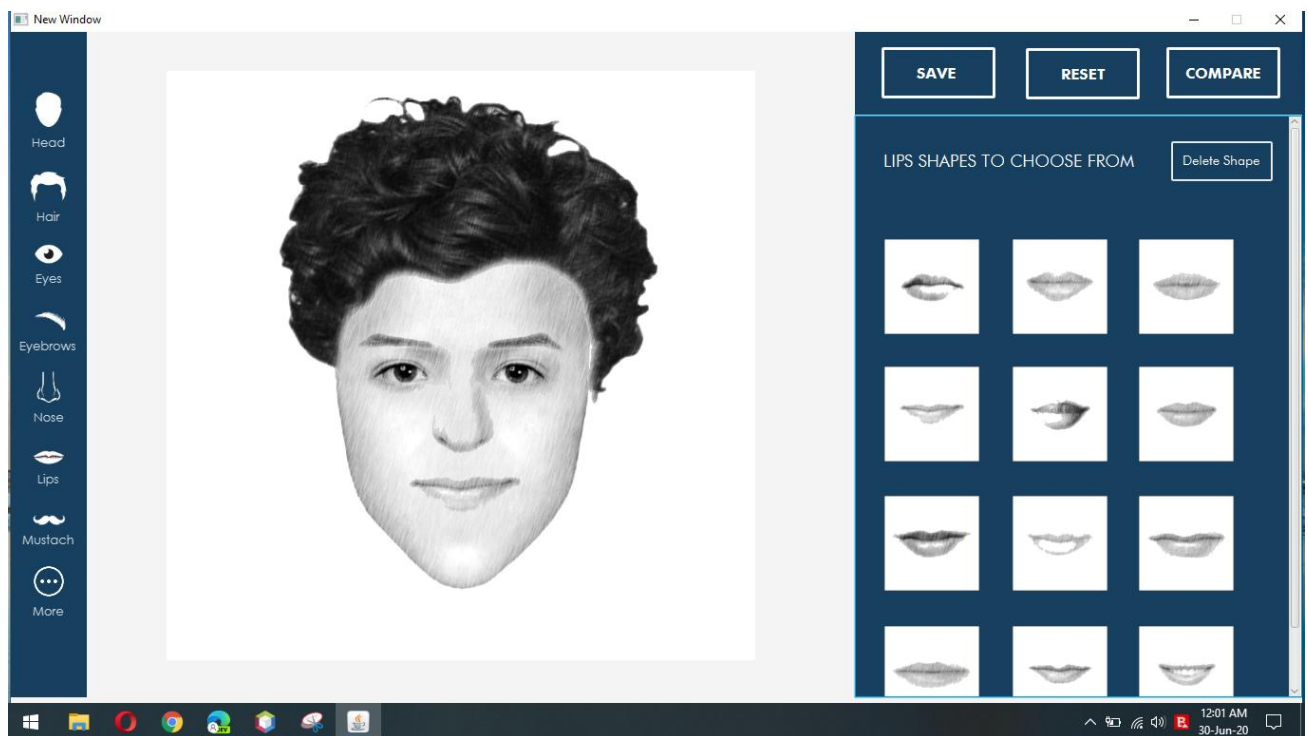


Fig 7.1.6 Face sketch using Drag and drop

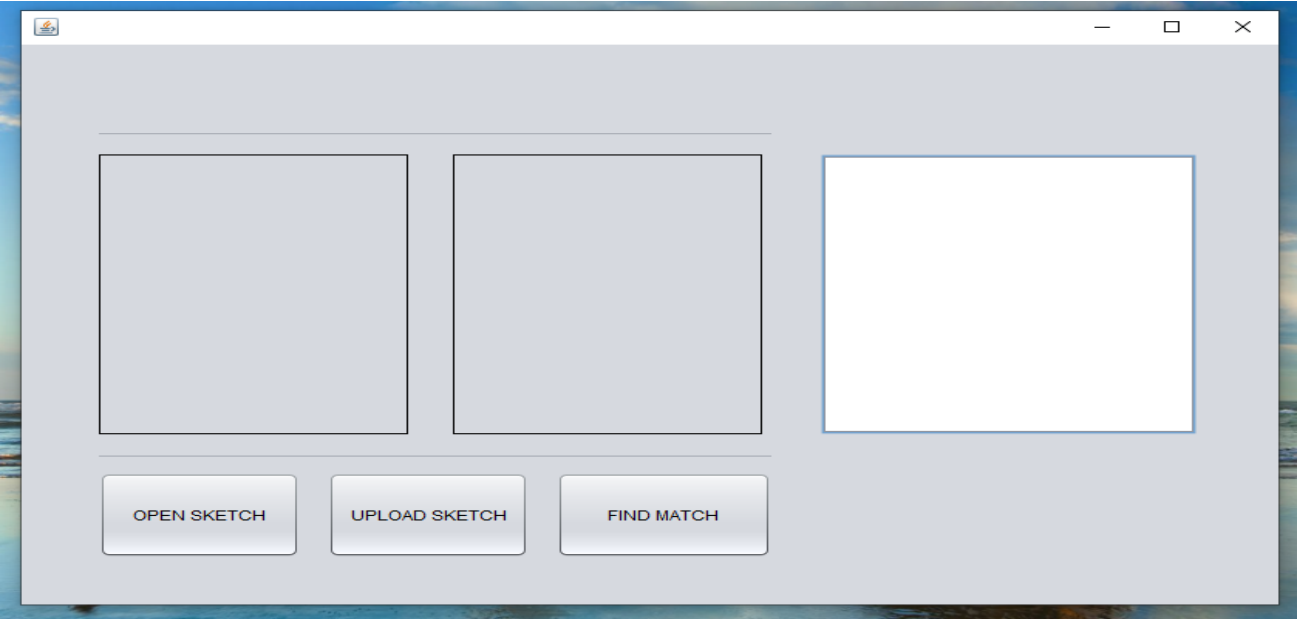


Fig 7.1.7 Recognition interface

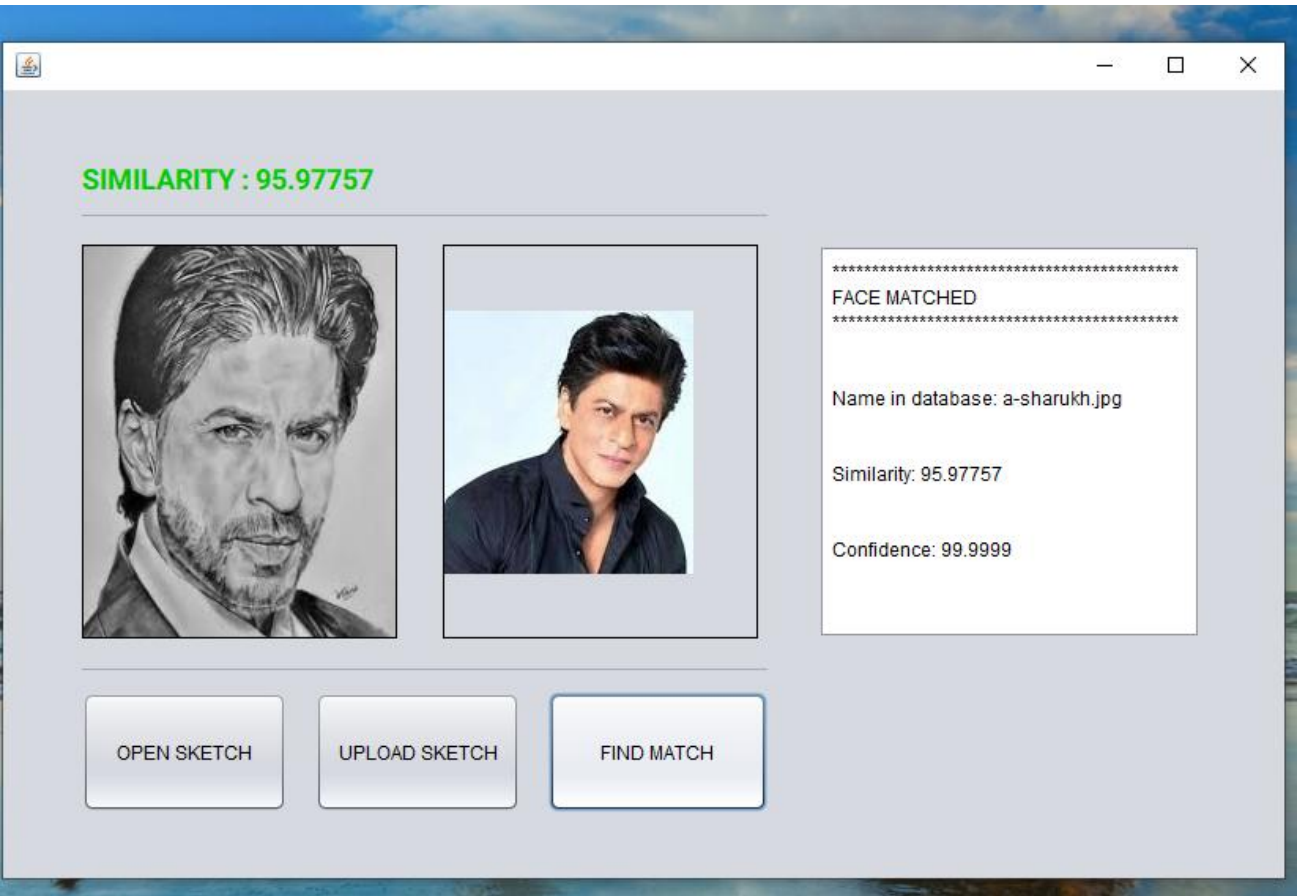


Fig 7.1.8 Face recognition

7.2 Future Augmentations & Refinements

The Third Eye system has strong foundational features for face sketch construction and recognition, but several enhancements can further improve its usability, accuracy, and real-world effectiveness. The following future augmentations and refinements can be considered for upcoming versions of the project:

1. Integration of Deep Learning–Based Sketch Generation

Future versions can incorporate AI-driven sketch generation where the system automatically creates a refined sketch based on descriptive text or a rough drawing. Natural Language Processing (NLP) can be used to convert eyewitness verbal descriptions into digital sketches.

2. Real-Time Facial Reconstruction from Voice Inputs

The system can be enhanced to allow eyewitnesses to verbally describe the suspect while the system automatically adjusts facial components in real-time using speech-to-feature mapping technology.

3. Support for 3D Face Modelling

A 3D model of the suspect’s face could be generated from the sketch, providing investigators with a more realistic and rotatable facial representation.

This would help when identifying suspects from different angles.

4. Advanced Sketch-to-Photo Domain Adaptation

Better deep learning models can be integrated to reduce the gap between hand-drawn sketches and real photographs.

Techniques like GANs (Generative Adversarial Networks) can enhance recognition accuracy by converting sketches into photorealistic images before matching.

5. Mobile Application Version

A mobile app can be developed to allow field officers to create and match sketches on the go. This would increase usability and support faster suspect identification during investigations.

6. Multi-Sketch Comparison

Future enhancements could allow investigators to upload multiple eyewitness sketches and merge them into a single composite sketch using AI-powered averaging and refinement.

7. Integration with National/Global Criminal Databases

The system can be expanded to connect with centralized police or international criminal databases (e.g., INTERPOL), enabling broader search capabilities and improving identification success rates.

8. Improved Drag-and-Drop UI with Gesture Controls

Advanced UI features, such as touch gestures, zooming, ultra-smooth alignment, and automatic proportional placement of components, can make the sketch creation process more intuitive.

9. Accuracy Feedback System

A feedback system could be added where investigators rate the accuracy of matches. This data can be used to continuously retrain the recognition model for better performance.

10. Automated Facial Component Suggestion

The system can analyse the partial sketch and suggest complementary facial features—for example, recommending suitable nose shapes or eye styles based on the structure already built.

11. Enhanced Security & Encryption

Future versions can include advanced encryption algorithms, facial authentication, and secure cloud hosting to safeguard sensitive criminal data.

12. Emotion & Age Estimation

The system can be extended to interpret and modify sketches based on estimated age progression, emotions, or disguises, which is useful when suspects attempt to alter their appearance.

CONCLUSION

The Third Eye – Face Sketch Construction and Recognition System successfully demonstrates how modern technologies can significantly enhance traditional forensic sketching and suspect identification methods. By integrating digital sketch construction, JavaFX-based interactive interfaces, image preprocessing, and machine learning–based face recognition, the system provides a reliable, efficient, and user-friendly platform for investigators. Unlike manual sketching, which relies heavily on artistic skill and subjective interpretation, the proposed system ensures a more standardized and consistent process through predefined facial components, drag-and-drop controls, and automated sketch rendering. This reduces human error and speeds up the investigative workflow, making the entire process more accessible and adaptable for law enforcement agencies.

Through the combination of OpenCV, deep learning models, and similarity-based matching algorithms, the system enables accurate sketch-to-photo recognition by converting sketches into numerical embeddings and comparing them against stored facial data. The implementation of preprocessing techniques such as grayscale conversion, normalization, and edge enhancement ensures that even rough or incomplete sketches can be analyzed effectively. With a structured database for storing criminal images, embeddings, and metadata, the system maintains organized data management and supports quick retrieval of potential matches. This creates a more transparent and automated identification pipeline that improves accuracy, boosts investigative speed, and supports decision-making.

Moreover, the use of JavaFX and Scene Builder provides an intuitive graphical user interface, ensuring smooth sketch creation and seamless navigation for users with varying technical backgrounds. By including modules for security, error handling, logging, and result reporting, the system emphasizes both reliability and accountability.

Overall, the project demonstrates the potential of combining computer vision, machine learning, and digital sketching to assist forensic investigations. It addresses major limitations of manual sketching, provides faster identification capabilities, and delivers a scalable and flexible solution that can be further expanded in the future.

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